



**Be careful what you pay for: The effect of performance
contingent incentives on online product reviews**

Journal:	<i>Journal of Research in Interactive Marketing</i>
Manuscript ID	JRIM-04-2026-0253.R2
Manuscript Type:	Research Article
Keywords:	e-commerce, online retailing, consumer behavior, word-of-mouth marketing



1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Be careful what you pay for: The effect of performance contingent incentives on online product reviews

Abstract

Purpose This research investigates how performance-contingent incentivization schemes affect online review contributions. It explores the interplay between financial incentives, prosocial reinforcement (social versus reciprocal exchange), and reviewer experience.

Design/methodology/approach The research conducts a field study in collaboration with a major North American online retailer to evaluate customer review behavior.

Findings Reviewer experience moderates the effect of performance-contingent incentives on review decisions and length. First-time (vs. experienced) reviewers see significant improvements in both review decisions and length with performance-contingent incentives. Additionally, because performance-contingent incentives reinforce reciprocal exchange, introducing reciprocal exchange reinforcement through messaging increases the level of effort but reduces review decisions. The optimal strategy depends on the platform’s proportion of first-time reviewers and the retailer’s specific review outcomes.

Originality This work identifies *reviewer experience* and *prosocial framing* as critical boundary conditions for the effectiveness of performance-contingent incentives.

Introduction

Online product reviews are important for consumers and retailers (Zhu *et al.*, 2018). Consumers expect a large number of reviews for every product (Chevalier, 2022). The majority of consumers read product reviews before buying a product and trust online reviews as much as personal recommendations (Pitman, 2022). Moreover, higher review volumes lead to more sales (Duan *et al.*, 2008; Ghose and Ipeirotis, 2010). Therefore, efforts must be made to attract review contributions cost-efficiently while encouraging reviewers to invest effort in what they write.

Accordingly, many retailers have adopted different incentivization schemes to entice their customers to write reviews (López *et al.*, 2022; Zhang *et al.*, 2024). Prior research has investigated different ways to better incentivize reviewers to write more substantive reviews. Some platforms utilize non-monetary reward schemes, such as “top reviewer badge” (Babić

Rosario *et al.*, 2020), while others use different monetary incentives, such as store credit (Burtch *et al.*, 2018) and cash rebates (Cabral and Li, 2015).

A key variable is the effort invested in writing online product reviews because it shapes how much information a review provides and whether the platform can sustain a helpful stock of user-generated content (Jia *et al.*, 2022; Li *et al.*, 2023). While incentivized reviews are beneficial for increasing the volume of reviews, they can reduce reviewer effort (evidenced by shorter reviews; Khern-am-nuai *et al.*, 2018; Ai *et al.*, 2022; Burtch *et al.*, 2018; Qiao and Rui, 2023). As a result, it is important to address the detrimental effect of financial incentives on review participation and effort.

A critical managerial pain point concerning review incentives remains largely unsolved. Standard incentive schemes that are contingent solely on writing a review (completion-contingent) are highly inefficient at converting the vast majority of a retailer's customer base to write reviews. An analysis of baseline data from a collaborating online retailer, which utilized a standard completion-contingent incentive scheme, highlights this inefficiency. While the retailer's experienced reviewers (42% of the customers) exhibited a relatively high participation rate (48.8%), the participation rate among the first-time reviewers (58% of the customers) was substantially lower (2.2%). Furthermore, because the reward was guaranteed regardless of the review effort, the resulting reviews were generally short. This creates two key problems for managers. The review budget is disproportionately spent on a smaller segment of existing reviewers and on unhelpful reviews (see Figure 1 for an illustration of this managerial tension).

— insert Figure 1 about here —

To solve this, retailers are utilizing performance-contingent incentives, which reward reviewers only if their reviews satisfy a platform-defined performance metric (Yu *et al.*, 2022). Performance-contingent incentives can increase review volume, length, and helpfulness for experienced reviewers (Dorner *et al.*, 2020). However, it remains unclear whether performance-contingent incentives can solve the participation bottleneck for the first-time reviewers (i.e., the majority of customers) who ignore review invitations.

Drawing on Self-Determination Theory (SDT) and Cognitive Evaluation Theory (CET; Deci *et al.*, 1999; Ryan and Deci, 2000), to align with emerging perspectives in interactive marketing (Wang, 2024), we demonstrate that performance-contingent incentives are effective means to encourage first-time reviewers to write reviews and increase their effort in writing reviews. By disentangling the psychological mechanisms behind review generation, this research answers calls to demonstrate theoretical contributions through clear behavioral storytelling (Wang, 2025). Furthermore, we identify the language utilized by retailers to solicit reviews as a critical boundary condition that interacts with these incentives. By manipulating the primary beneficiary of the review as the company (vs. other customers; Wadi *et al.*, 2026), we show that for first-time reviewers, reciprocal framing lowers overall participation but isolates a subset of customers heavily driven to reaffirm their competence. As a result, introducing a performance-contingent incentive paired with reciprocal framing yields the highest relative increase in review effort from first-time reviewers.

To test our hypotheses, we operationalize a performance-contingent incentive scheme by fixing the total incentive budget. Following a scenario-based calibration study (detailed in Web Appendix A) to determine the optimal payout ratio, we conducted a 2 (Performance-

contingency: Yes vs. No) x 2 (Prosocial framing: Reciprocal vs. Social) between-subjects quasi-experiment with a major North American online retailer. Sending 9,555 review invitations, which yielded 1,660 reviews, we tested the main and interaction effects of incentive type, reviewer experience, and prosocial framing on both the decision to write a review and review length.

Our findings make three theoretical contributions. First, we advance the understanding of financial incentives and user-generated content (Burtch *et al.*, 2018) by conceptually disentangling the *informational* versus *controlling* aspects of performance-contingent incentives (Dorner *et al.*, 2020; Yu *et al.*, 2022). Second, we identify critical boundary conditions of incentive effectiveness. We demonstrate that a reviewer's baseline experience moderates their response to incentive structures, where first-time reviewers exhibit the strongest positive effort response to performance-contingent incentives. Third, we provide a more nuanced understanding of online review generation by showing that the psychological process governing the decision to participate differs from the process governing task effort. By showing how prosocial framing interacts with these stages differently, we extend findings on exchange motives in interactive marketing (Wadi *et al.*, 2026).

Practically, our findings provide managers with concrete, actionable guidelines for devising effective incentivization schemes that optimize for review helpfulness, cost per helpful review, and review conversion rate. We devise a data-driven decision tree that allows managers to determine the optimal incentive scheme (completion-contingent vs. performance-contingent) and framing strategy (social vs. reciprocal exchange) based on the composition of their platform's customer base (i.e., the percentage of first-time reviewers).

Literature Review

Online product reviews, a form of electronic word-of-mouth (eWOM; Hennig-Thurau *et al.*, 2004; Babić Rosario *et al.*, 2020), are essential for the operational success of retailers (López *et al.*, 2022), yet two interconnected challenges persist: generating a high volume of reviews and encouraging reviewers to invest substantive effort into their reviews (Burtch *et al.*, 2018; Khernam-nuai *et al.*, 2018).

To overcome these challenges, retailers have utilized various incentive schemes to increase review volume, including free products, discounts, and financial rewards (Greene, 2021; Smith, 2021; Woolley and Sharif, 2021). Completion-contingent financial incentives are prevalent in the industry and have been the primary focus of academic research. In a completion-contingent scheme, reviewers receive a guaranteed financial reward simply for submitting a review, regardless of its quality or helpfulness to other consumers (Yu *et al.*, 2022). While completion-contingent incentives are effective at eliciting large volumes of reviews (Klein *et al.*, 2018), they fail to promote review effort. Research demonstrates that providing completion-contingent financial incentives decreases review length, linguistic effort, and overall depth of the resulting reviews (Burtch *et al.*, 2018; Qiao *et al.*, 2020). As a result, there is a clear need for incentive designs that preserve review participation *and* effort.

Performance-contingent incentive schemes have been proposed as a solution to improve the effort and helpfulness of user-generated content (Dorner *et al.*, 2020; Wang *et al.*, 2012; Yu *et al.*, 2022). Unlike completion-contingent incentives, performance-contingent incentives provide rewards only if the review meets specific performance requirements. Research shows that in incentivized settings, consumers often perform poorly when compensation is completion-

contingent because the incentive is perceived merely as payment for their participation, whereas tying rewards to performance metrics can increase task effort (Gneezy and Rustichini, 2000).

Research has examined performance-contingent incentives through winner-takes-all tournament structures, such as the Amazon Vine program (Dorner *et al.*, 2020; Yu *et al.*, 2022). In these competitive schemes, a select group of highly experienced reviewers compete for a limited pool of rewards contingent on performance. While this increases review quantity among elite reviewers, it fails to improve overall review quality, encourages strategic downvoting by peers, and diminishes the signaling power of helpfulness ratings (Dorner *et al.*, 2020; Yu *et al.*, 2022).

Prior literature has focused mainly on top reviewers or tournament winners (Dorner *et al.*, 2020; Yu *et al.*, 2022). Therefore, we know little about the behavioral impact of performance-contingent incentives when offered to the broader customer base. This gap is particularly critical regarding first-time reviewers (i.e., customers who have never written a review because they ignore standard completion-contingent invitations). Furthermore, prior research has largely overlooked how the linguistic framing of these incentive invitations (e.g., social vs. reciprocal language) interacts with the reward structure to shape reviewer behavior, despite recent evidence that exchange framing significantly impacts online reviews (Burtch *et al.*, 2018; Wadi *et al.*, 2026).

In this research, we address these gaps by investigating a fixed-budget, non-competitive performance-contingent incentive scheme offered to the broad customer base. Furthermore, introducing this scheme to a platform currently utilizing completion-contingent incentives allows us to isolate the differential effects of performance-contingent incentives on both first-time and experienced reviewers.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Theoretical framework and hypothesis development

Figure 2 shows our theoretical framework.

— Figure 2 around here —

Self-Determination Theory and Cognitive Evaluation Theory

To conceptualize the psychological mechanisms underlying online review behavior, we draw on Self-Determination Theory (SDT), a prominent macro-theory of human motivation (Deci *et al.*, 1999; Gilal *et al.*, 2019; Ryan and Deci, 2000). SDT postulates that human motivation is governed by three basic psychological needs: *autonomy* (the desire to experience behavior as volitional and self-endorsed), *competence* (the desire to feel effective and capable in one’s actions), and *relatedness* (the desire to feel connected to others). In the context of interactive marketing, the fulfillment of these needs fosters autonomous motivation, which is a primary driver of user participation and content generation (Nguyen and Hess, 2026; Wang and Li, 2014).

It is necessary to examine how external rewards (e.g., financial incentives) interact with these basic psychological needs. Cognitive Evaluation Theory (CET), a sub-framework within SDT, provides a theoretical basis for this dynamic (Rummel and Feinberg, 1988). According to CET, external rewards do not uniformly impact motivation (Ryan *et al.*, 1983). Rather, they possess a dual nature, containing both a *controlling aspect* and an *informational aspect* (Deci *et al.*, 1999; Kim and Ahn, 2017).

The *informational* aspect of a reward refers to its capacity to provide competence-affirming feedback (Vallerand and Reid, 1984). When a reward signifies that an individual has

successfully met an established performance standard, it serves as a tangible signal of competence, thereby motivating individuals to allocate cognitive resources toward the task to attain validation (Mekler *et al.*, 2017). In contrast, the *controlling* aspect of a reward reflects the degree to which an external contingency pressures an individual to meet a specific outcome. When rewards are conditional on performance metrics, they shift the perceived locus of causality from internal to external (Ryan and Deci, 2000), which thwarts the individual's need for autonomy and decreases the willingness to participate (Gneezy *et al.*, 2011).

Thus, by applying the CET framework to the design of online review incentives, we conceptually separate reviewer behavior into two distinct processes. First, the controlling aspect of an incentive structure primarily influences the decision to participate in the task (i.e., the decision to write a review). Second, for those who choose to participate, the informational aspect of the incentive structure influences the subsequent resources allocated to the task (i.e., the review effort).

Reviewer experience

Within the CET framework, an individual's response to external rewards is moderated by their existing level of competence (i.e., baseline competence) regarding the task (Chen, Zhang, *et al.*, 2023; Deci *et al.*, 1999; Vallerand and Reid, 1984). In the context of an online retail platform, a user's baseline competence can be measured by their prior review experience (Huang *et al.*, 2015; Racherla and Friske, 2012). Experienced reviewers (i.e., customers who have previously authored reviews on the platform) possess a relatively higher level of baseline competence compared to first-time reviewers. Through prior engagement, they have developed established routines and familiarity with the platform's evaluation mechanisms. Because their competence is already established, the perceived difficulty of the task is comparatively low, and

they have a higher baseline autonomous motivation (Choi and Maasberg, 2022). Therefore, experienced reviewers are naturally predisposed to exhibit higher participation rates and invest more effort in their reviews than first-time reviewers (Moe and Schweidel, 2012).

H1 Experienced (vs. first-time) reviewers **(a)** demonstrate a higher probability of writing a review and **(b)** write longer reviews.

The introduction of performance-contingent incentives alters this baseline behavior, but the magnitude and direction of the effect depend on how reviewer experience moderates the dual aspects of the reward (i.e., the controlling versus informational aspects).

Regarding the decision to write a review, the behavior is governed by the controlling aspect of the incentive. Because experienced reviewers have high baseline autonomous motivation, transitioning from a guaranteed completion-contingent scheme to a performance-contingent scheme imposes external constraints on their behavior. This controlling aspect of the reward crowds out their existing autonomous motivation (Frey and Jegen, 2001; Krishna *et al.*, 2024), resulting in a negative shift in their likelihood to participate. First-time reviewers, on the other hand, lack this baseline autonomous motivation. Therefore, there is little autonomous motivation to crowd out, making them less negatively affected by the controlling aspect of the performance-contingent requirement (Goette and Stutzer, 2020). Therefore, the effect of performance-contingent incentives on task participation is relatively more positive for first-time reviewers compared to experienced reviewers.

H2.a Performance-contingent (vs. completion-contingent) incentives have a more positive effect on the probability of writing a review for first-time reviewers compared to experienced reviewers.

Conversely, once the decision to participate is made, task effort is governed by the informational aspect of the incentive. Experienced reviewers already possess high perceived competence. Therefore, they are less dependent on the informational feedback of a performance-contingent reward to validate their competence. As a result, their effort remains relatively unaffected.

In contrast, for first-time reviewers, the informational aspect of a performance-contingent reward is highly salient because it offers a tangible opportunity to establish competence on the platform (Finkelstein and Fishbach, 2012). The mechanism linking this salience to task effort follows directly from CET's competence-feedback logic. Because first-time reviewers lack baseline competence, external performance-contingent feedback is diagnostic for them. It is an available means of determining whether they are capable of performing the task effectively (Deci *et al.*, 1999; Vallerand and Reid, 1984). According to CET, individuals driven to satisfy a need for competence actively allocate greater cognitive resources to a task to secure that validation (Gagné and Deci, 2005). Effort is therefore instrumental. Since satisfying the performance metric (e.g., securing a helpful vote) requires a more effortful review, exerting greater effort is a primary means by which first-time reviewers can obtain the competence-affirming feedback they seek (Mekler *et al.*, 2017). Accordingly, to maximize the probability of securing this validation, first-time reviewers increase their task effort (Goes *et al.*, 2014). Experienced reviewers, whose competence is already established and for whom such feedback carries little diagnostic value, have no corresponding motivation to increase effort.

H2.b Performance-contingent (vs. completion-contingent) incentives have a more positive effect on review length for first-time reviewers compared to experienced reviewers.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

The need for relatedness and prosocial framing

Beyond the structural design of the incentive, the language utilized by retailers to solicit reviews influences the reviewer’s behavior (Wadi *et al.*, 2026). Within the SDT framework, the third basic psychological need is *relatedness*, the innate desire to feel connected to and meaningfully helpful toward others (Deci *et al.*, 1999). Retailers can reinforce different psychological motives by framing the primary beneficiary of the review as either the company (reciprocal exchange) or other consumers (social exchange; Wadi *et al.*, 2026; Chen, Yuan, *et al.*, 2023; Rawal *et al.*, 2024).

When a retailer utilizes a reciprocal exchange framing (e.g., soliciting the review to “help the company” in exchange for a reward), it emphasizes the bilateral, economic nature of the interaction (Hennig-Thurau *et al.*, 2004). For first-time reviewers, who possess low autonomous motivation for writing a review, this transactional framing amplifies the *controlling aspect* of a performance-contingent reward. It creates an environment where the reviewer feels pressured to produce content that serves corporate interests, which thwarts their sense of autonomy and deters participation (Burtch *et al.*, 2018).

In contrast, a social exchange framing (e.g., asking the customer to “help other customers”) alters the psychological context by triggering the need for relatedness (Chen, Yuan, *et al.*, 2023). Focusing on prosocial contribution softens the controlling nature of performance-contingent incentives by activating altruistic motives and satisfying the basic psychological need for relatedness (Baumeister and Leary, 1995; Ryan and Deci, 2000; Wadi *et al.*, 2026; Weinstein and Ryan, 2010). By shifting the reviewer’s focus away from a strict corporate transaction and toward a community-oriented contribution, the task fosters greater autonomous motivation

(Weinstein and Ryan, 2010). Therefore, for first-time reviewers, social (vs. reciprocal) exchange framing lowers the psychological barrier to entry (Wadi *et al.*, 2026).

H3.a Reinforcing reciprocal (vs. social) exchange motives weakens the moderating effect of being a first-time (vs. experienced) reviewer on the relationship between performance-contingent (vs. completion-based) incentives and the probability of writing a review.

Since social (vs. reciprocal) exchange framing triggers the need for relatedness and lowers the psychological barrier to entry, it invites a broader, more heterogeneous pool of first-time reviewers to write reviews. This broader pool encompasses both individuals seeking the informational aspect to reaffirm their competence, as well as those primarily driven by the need for relatedness and the intrinsic desire to help others, for whom the informational aspect is secondary. Because relatedness-driven individuals are less focused on maximizing the external performance metric, the effort within this expanded pool is diluted (Bénabou and Tirole, 2004; Burtch *et al.*, 2018; Khern-am-nuai *et al.*, 2018).

In contrast, the controlling aspect of the reciprocal exchange framing acts as a filter that isolates a subset of first-time reviewers who are predominantly driven by the informational aspect of the reward (Cadsby *et al.*, 2007). Because this pool is concentrated with individuals seeking to reaffirm their competence, they exert higher effort to ensure they meet the platform's performance metric (Eisenberger *et al.*, 1999). Therefore, first-time reviewers operating under reciprocal (vs. social) exchange framing exert higher effort to secure the competence-affirming, performance-contingent incentives.

H3.b Reinforcing reciprocal (vs. social) exchange motives strengthens the moderating effect of being a first-time reviewer (vs. experienced) on the relationship between performance-contingent (vs. completion-based) incentives and review length.

To test our hypotheses, we conducted a field study with a major online retailer and observed purchase and review behaviors.

Field study

We carry out a field study to test our hypotheses (H1-H3) in a natural setting where actual reviews are written, and the length of the reviews can be used to measure review effort. Prior to the field experiment, we conducted a scenario-based calibration study (detailed in Web Appendix A) to determine the optimal proportion of performance-contingent incentives to deploy in the field. Results indicated that an 80% performance-contingent ratio yields the optimal balance of review intention and expected effort, which we subsequently adopted for our main field intervention.

Experiment design

We study the effects of performance-contingent (vs. completion-contingent) incentives and prosocial reinforcement in a 2 (Performance-contingency: Yes vs. no) x 2 (Reciprocal exchange reinforcement: Yes vs. No) between-subjects design. We collaborated with a major North American online retail store. The retailer already incentivizes its reviewers by sending them a review solicitation message after each purchase and offers them a \$10 store credit after their review is published on the retailer’s platform. Thus, we know that experienced reviewers have already received financial incentives in exchange for their reviews. More importantly, we

know that completion-contingent incentives are the norm. Therefore, experienced reviewers (i.e., those who have written at least one review) are used to completion-contingent incentives. Under this baseline completion-contingent program, without any intervention, platform data showed that experienced reviewers had a 48.8% response rate, whereas first-time reviewers had only a 2.2% response rate, demonstrating the platform's struggle to activate new reviewers.

For reinforcing reciprocal exchange we used a review solicitation message that framed the review as helping the company be successful. For this group, since the beneficiary of altruism is the provider of financial incentives, the economic exchange motives are reinforced (Wadi *et al.*, 2026). For the group with no reciprocal exchange reinforcement, we used a review solicitation message that framed the review as helping other customers. In this condition, since the beneficiary of altruism is a separate party from the provider of financial incentives, the social exchange motives are reinforced (Wadi *et al.*, 2026). Table I details the wording of the emails sent for each group.

To verify the manipulation of the social and reciprocal exchange framings, we conducted an independent manipulation check (see Web Appendix B for detailed results), which showed that the framing manipulations were successful.

Sample and Procedure

We sent the review solicitation emails for each condition in different time windows. We could not manipulate the number of emails sent for each group since it depends solely on the number of orders placed on each particular day. We collected the review data 91 days after the first review solicitation email was sent. Thus, each review solicitation email had between 85 and 91 days to lead to a review decision and receive helpful votes. In some cases, a review was due

to a reminder email, which was sent after the treatment period and did not include the manipulated review solicitation message. Therefore, we excluded those reviews from all groups to compile the set of valid emails (Table I). Additionally, for all groups, we told the customers that the financial incentive would expire 90 days after it was granted. Thus, for the performance-contingent groups, the reviewer had 90 days to receive a helpful vote to receive and use the performance-contingent incentive. This resolves technical and legal issues present in other performance-contingent schemes (Wang *et al.*, 2012).

For the performance-contingent groups, the retailer told their customers that they would receive \$2 in store credit after their review was published (completion-contingent) and \$8 in store credit after their review received a helpful vote (performance-contingent). For the completion-contingent groups, the retailer told their customers that they would receive \$10 in store credit after their review was published (the retailer’s default incentive scheme). For this group, no incentive for receiving a helpful vote was offered.

— Table I around here —

Measures

Because our primary independent variables are randomly assigned experimental conditions, they are operationalized as manipulated binary dummy variables based on the specific email treatment the customer received (see Table I). Performance-contingency is coded as 1 for reviewers who received the performance-contingent incentive and 0 for those who received the completion-contingent incentive. Reciprocal exchange reinforcement is coded as 1 for reviewers who received the reciprocal exchange message framing and 0 for those who received the social exchange message framing. To ensure the scientific validity and reliability of

these experimental manipulations, we conducted independent manipulation checks (detailed in Web Appendix A and Web Appendix B), which confirm that our treatment designs isolated the intended incentive structures and prosocial motives.

Prior research has used review length to measure reviewer effort (Rohde *et al.*, 2022). Importantly, longer reviews are associated with more helpfulness (Jia *et al.*, 2022). To empirically validate this relationship within our specific dataset, we conducted a robustness check (detailed in Web Appendix D), which confirms that review length significantly predicts the likelihood of receiving a helpful vote.

We focus on review length rather than helpfulness votes as our primary dependent variable for two critical reasons. First, from a theoretical standpoint, review length is a proximal and direct measure of the reviewer's actual effort. While reviewers have immediate control over how much effort they exert in writing a review, they have no direct control over the subsequent market outcomes (i.e., whether another consumer finds their review helpful). Second, from a statistical perspective, helpfulness votes in our dataset are highly zero-inflated (historically, only 3.9% of reviews on the retailer's platform had received at least one helpful vote), which severely limits the statistical power required to detect nuanced interaction effects in a field experiment. We, therefore, treat helpful votes as secondary outcomes that reflect reader response rather than a primary measure of reviewer effort. Review length remains our central dependent variable because it is continuous, reviewer-controlled, and directly aligned with our effort-based theoretical grounding. We measured helpfulness of the reviews 90 days after the review solicitation emails were sent. Helpfulness is measured by subtracting thumbs-down votes (if any) from the thumbs-up votes of the review.

To account for contextual factors that may influence review behavior, we include several covariates. *Order day* represents the day of the week the order was placed. We include day-of-the-week fixed effects to account for potential temporal confounds arising from the sequential deployment of the quasi-experimental conditions. Because consumer purchasing and reviewing behaviors naturally fluctuate across the week (e.g., weekend versus weekday trends), daily order volumes varied across our IT deployment windows. These fixed effects absorb baseline temporal noise, such as weekend effects or existence of promotions on certain days. Thus, by controlling for *Order day*, we ensure that our estimated effects are driven by the incentive structures and framings rather than unobserved temporal variations.

Review score is operationalized via the 1–5 star rating given to the product, which controls for the tendency of consumers with extreme product experiences to write more detailed reviews. *Brand popularity* is a derived binary variable indicating whether the purchased brand is *popular* or *niche*. This derivation was motivated by the data sparsity from a long tail of 286 unique brands that were purchased. We sorted brands by their total review volume. Brands comprising the top 50% of the platform’s total cumulative reviews are coded as popular, while the remaining brands are coded as niche.

To measure reviewer experience, we derive *first-time reviewer* as a binary variable that takes the value 1 if the reviewer has not written any reviews on the platform, and 0 if they have (i.e., experienced reviewers). We deliberately avoid including continuous, overlapping measures of experience, such as historical review count or platform tenure, as covariates in our models. Because these variables are structurally collinear with our primary theoretical construct of reviewer experience (e.g., a first-time reviewer inherently has a historical review count of zero), including them both would introduce severe multicollinearity.

Finally, to ensure that the deployment of emails across different time windows did not introduce endogeneity due to varying customer profiles or product mixes, we conducted a formal randomization and balance check on key covariates. While product-related covariates were well balanced, we observed a statistically significant variation in the proportion of first-time reviewers across deployment windows. However, because reviewer experience is explicitly modeled as a moderating factor in all our analyses, our models account for this baseline heterogeneity without biasing the treatment estimates (see Web Appendix C for detailed results).

Results

Data processing and statistical modeling were conducted using Python (version 3.13.12) and R (version 4.6.1). To model review length as a function of performance-contingency, reciprocal exchange reinforcement, and first-time reviewer, we use the Box-Cox method (*MASS* package in R) to find the appropriate transformation parameter, λ (Box and Cox, 1964). This mathematical transformation ensures the scientific reliability of our linear models by correcting for the positive skewness inherent in count data. The Box-Cox test shows square root is the optimal transformation for review length ($\lambda=0.22$).

To test our hypotheses, we estimated a logistic regression for the decision to write a review and an ordinary least squares (OLS) regression for the square root of review length using the *statsmodels* library (version 0.14.6) in Python. All models include day-of-the-week fixed effects to control for temporal variations in baseline reviewing behavior across the sequential deployment windows.

The results show a significant main effect of first-time reviewer on the review decision ($B=-2.91$, $z=-34.31$, $p<0.001$; Table III, Decision - Main) and the square root of review

length ($B=-0.91$, $t=-2.48$, $p=0.013$; Table III, Length - Main), indicating that experienced reviewers are likelier to write reviews (**H1.a** supported) and write significantly longer reviews (**H1.b** supported) than first-time reviewers.

The results show a significant two-way interaction between performance-contingency and first-time reviewers on review decision ($B=0.92$, $z=5.25$, $p<0.001$; Table III, Decision - 2-Way) and the square root of review length ($B=1.57$, $t=2.10$, $p=0.036$; Table III, Length - 2-Way). This confirms that performance-contingent incentives have a more positive impact on first-time (vs. experienced) reviewers for both task participation and effort (**H2.a** and **H2.b** supported).

— Table III around here —

Furthermore, the results show a significant three-way interaction among performance-contingency, reciprocal exchange reinforcement, and first-time reviewer on the decision to write a review ($B=-1.14$, $z=-3.18$, $p=0.001$; Table III, 3-Way; Figure 3; **H3.a** supported) and the square root of review length ($B=3.70$, $t=2.42$, $p=0.016$; Table III, 3-Way; Figure 4; **H3.b** supported). This indicates that when performance-contingent incentives are provided, first-time reviewers see the economic exchange paradigm reinforced through reciprocal exchange reinforcement as a major barrier to entry and react negatively to it. Those who pass this barrier, on the other hand, exert more effort.

Figure 3 shows that reinforcing social (vs. reciprocal) exchange increases review decisions for both first-time and experienced reviewers (Wadi *et al.*, 2026).

— Figure 3 around here —

For the review effort, Figure 4 (left) shows that when reciprocal exchange is reinforced, first-time reviewers who receive performance-contingent ($M_{length}=156.16$; $SE_{length}=20.05$) versus completion-contingent ($M_{length}=93.66$; $SE_{length}=17.97$) incentives write significantly longer reviews. Figure 4 (right) shows that when social exchange is reinforced, first-time reviewers who receive performance-contingent ($M_{length}=126.90$; $SE_{length}=19.71$) versus completion-contingent ($M_{length}=132.18$; $SE_{length}=25.50$) incentives do not show any significant increase in their review length. Furthermore, Figure 4 also shows that experienced reviewers do not show any significant change in their review length regardless of the prosocial reinforcement or performance-contingent incentive.

Finally, we note that while our experimental treatments produced statistically significant and managerially meaningful shifts in reviewer behavior, the overall explanatory power (R^2) of the review length models is relatively low (e.g., $R^2=0.020$). This is expected in large-scale field experiments analyzing noisy, individual-level human behavior (Wooldridge, 2016). The decision of exactly how many characters to write in a product review is driven by various unobservable idiosyncratic factors (e.g., the reviewer's daily schedule, mood, or writing ability) that are orthogonal to our experimental manipulation. Because our objective is causal inference (i.e., isolating the treatment effect of performance-contingent incentives and prosocial reinforcement) rather than building a predictive model of review length, these low R^2 values do not diminish the validity or significance of the estimated treatment effects.

— Figure 4 around here —

Post-hoc analysis

We perform a simulation on the data using the *NumPy* library (version 2.4.4) in Python to optimize several managerial outcomes of interest. Since experienced and first-time reviewers react differently to different review incentivization schemes and prosocial reinforcements, we simulate conversion rate, helpful review (%), and cost per helpful review (\$) for different percentages of first-time reviewers on a given platform. We focus on review helpfulness because research has shown that review helpfulness is an important predictor of product sales (Lee and Choeh, 2020), making it important for managerial decisions. We measure *conversion rate* as the percentage of the total review solicitation messages sent by the retailer that led to a review. *Helpful review (%)* shows the percentage of helpful reviews to the total number of reviews written. *Cost per helpful review (\$)* measures the total incentive (performance-contingent and completion-contingent) that the retailer should pay to receive one helpful review.

Using the two experimental factors (Performance-contingency: Yes vs. no x Reciprocal exchange reinforcement: Yes vs. no), we constructed four incentive schemes and prosocial reinforcements groups (Completion-contingent/Social exchange reinforcement, Performance-contingent/Social exchange reinforcement, Completion-contingent/Reciprocal exchange reinforcement, and Performance-contingent/Reciprocal exchange reinforcement). Next, we calculated the conversion rate, helpful review (%), and cost per helpful review (\$) for first-time and experienced reviewers within each group from the data. Then we simulated for different percentages of first-time reviewers (i.e. 0% to 100%) the overall conversion rate, helpful review (%), and cost per helpful review (\$), by taking the linear combination for each group's first-time and experienced reviewers.

For retailers interested in maximizing their conversion rate, Figure 5 shows that the highest achievable outcome at any point on the graph is always obtained by reinforcing social exchange. In terms of incentive scheme, if less than 65% of the review platform consists of first-time reviewers, a completion-contingent incentive is the optimal scheme, otherwise a performance-contingent incentive leads to a higher conversion rate.

— Figure 5 around here —

For retailers who want to maximize the percentage of helpful reviews on their platform, Figure 6 shows that a performance-contingent incentive is the optimal incentive scheme, regardless of the percentage of first-time reviewers. Platforms with under 34% first-time reviewers should use reciprocal exchange reinforcement in their review solicitation messages, because for these platforms experienced reviewers are the main determinants of helpful votes, and they are driven by established reciprocal motives. Social exchange should be reinforced for platforms with over 34% first-time reviewers. The data shows that when performance-contingent incentives are offered, even though first-time reviewers have a lower average review length when social (vs. reciprocal) exchange is reinforced (153.54 vs. 192.34 characters; Table II), reinforcing social exchange leads to higher average percentage of helpful votes than reciprocal exchange (11.27% vs. 8.86%). This divergence highlights the distinction between raw effort in performing the task versus the market's perception of its usefulness. It is important to note that these metrics are descriptive aggregates used to inform the managerial simulation rather than formal statistical tests. However, the pattern suggests that specific framings may alter how the market receives that effort.

— Figure 6 around here —

Finally, for retailers who wish to minimize the cost per helpful review, Figure 7 shows that completion-contingent incentive schemes ($M_{cost}=\$201.89$, $SD_{cost}=\$25.16$) are far more expensive than performance-contingent schemes ($M_{cost}=\$37.24$, $SD_{cost}=\$5.76$). The optimal prosocial reinforcement if the percentage of first-time reviewers is less than 64% is reciprocal exchange. In contrast, social exchange is optimal if the percentage of first-time reviewers is greater than 64%.

— Figure 7 around here —

Discussion

We used a field study with a retailer that incentivizes all of its customers in a completion-contingent scheme. Therefore, for experienced reviewers, we know that they have already entered a reciprocal exchange paradigm by writing reviews in exchange for financial incentives. We found that experienced reviewers have a higher probability of writing a review (H1.a) and write longer reviews (H1.b) than first-time reviewers. These findings confirm the need to improve the review decision and length for first-time reviewers.

We found that first-time (vs. experienced) reviewers are more positively affected by performance-contingent incentives, demonstrating a larger increase in their review decisions (H2.a) and length (H2.b) when they receive performance-contingent incentives. It is important to note that the performance metric used in this context relies on a weak signal: historically, only 3.9% of reviews on this platform received a helpful vote. This low probability of reward realization provides further context for the moderating role of reviewer experience. Experienced reviewers likely possess accurate expectations about the difficulty of achieving a helpful vote, which may contribute to their lack of effort response to the performance-contingent incentive. In

contrast, first-time reviewers, lacking this historical platform knowledge, exert greater effort in anticipation of a payoff they perceive as attainable. As a result, a performance-contingent incentive is an effective method for tackling the issue of low review decision and length for first-time reviewers.

The results also showed that reinforcing social (vs. reciprocal) exchange motives strengthens the effect of performance-contingent (vs. completion-contingent) incentives on the review decisions for first-time (vs. experienced) reviewers (H3.a), by increasing their sense of relatedness. On the other hand, we found that reinforcing reciprocal (vs. social) exchange motives increases the effect of performance-contingent (vs. completion-contingent) incentives on the review length for first-time (vs. experienced) reviewers (H3.b), by self-selecting the reviewers who are primarily driven by competence-affirming motivation. As a result, our moderated-moderation approach demonstrates that reciprocal (vs. social) framing is a boundary condition that shapes how performance-contingent incentives affect review effort and length.

In a post-hoc analysis, we executed a simulation to find the optimal strategies for maximizing the conversion rate and the percentage of helpful reviews and minimizing the cost per helpful review (Figure 8). In a prescriptive analysis, we showed that the optimal incentive scheme for maximizing helpful review percentage and minimizing cost per helpful review is performance-contingent incentives. The only scenario where the retailer should use completion-contingent incentives is when it is interested in maximizing its conversion rate and the percentage of first-time reviewers on its platform is less than 65%.

We also showed that to achieve the optimal conversion rate, helpful review percentage, and cost per helpful review, retailers should choose the right type of prosocial reinforcement in their review solicitation message. If the retailer is interested in maximizing its conversion rate, it

should reinforce social exchange, regardless of the percentage of first-time reviewers on its platform. The comparative advantage of social exchange reinforcement over reciprocal exchange reinforcement in increasing review decisions has been previously shown (Wadi *et al.*, 2026). Finally, we found that when the percentage of first-time reviewers is higher than 34% and 64%, the retailer should reinforce social exchange to maximize helpful review percentage and minimize cost per helpful review. Otherwise, it should reinforce reciprocal exchange because the effect is primarily driven by experienced reviewers and established reciprocal motives.

Theoretical Contributions

Our work makes three primary theoretical contributions to the literature on user-generated content and performance-contingent incentives (Dorner *et al.*, 2020; Yu *et al.*, 2022) by unifying these phenomena under Self-Determination Theory (SDT) and Cognitive Evaluation Theory (CET).

First, we extend the application of incentive design in online reviews. While prior literature has predominantly focused on the binary presence or absence of financial rewards (Burtch *et al.*, 2018), we apply CET to conceptually disentangle the dual nature of these incentives. We demonstrate that the controlling aspect of a performance-contingent incentive governs the initial decision to participate, whereas the informational aspect independently governs the subsequent effort exerted on the task.

Second, we identify reviewer experience as a critical boundary condition of incentive effectiveness through baseline competence. Standard economic intuition suggests that novice users, lacking experience, should be risk-averse and retreat from strict performance-based contracts. However, viewed through the lens of SDT, we demonstrate a counter-intuitive

phenomenon: because first-time reviewers lack baseline competence on the platform, the informational feedback of a performance-contingent incentive drives them to write more reviews and exert more effort in doing so. Experienced reviewers, whose competence is already established, show no significant improvement.

Third, we advance the literature on prosocial motives in electronic word-of-mouth (Wadi *et al.*, 2026) by demonstrating how linguistic framing interacts with the informational and controlling aspects of financial rewards. We reveal a theoretical trade-off between task participation and task effort. While social exchange framing fulfills the psychological need for relatedness and lowers the barrier to participation, it dilutes overall task effort. In contrast, reciprocal exchange framing amplifies the controlling aspect of the reward, acting as a filter that lowers overall participation but isolates a subset of users highly driven to reaffirm their competence, thereby maximizing task effort.

Managerial Implications

Many retailers have allocated substantial budgets to incentivizing reviewers, yet these budgets are frequently spent on the wrong reviewers and the wrong reviews. Our findings speak directly to three decisions managers face when designing a review incentive scheme: (1) who to target (experienced vs. first-time reviewers), (2) what to condition payment on (completion vs. performance), and (3) what to say in the solicitation message (help the company vs. help other customers). We organize our implications around these three decisions and quantify the business impact of each.

To illustrate the practical implications of our findings, consider the baseline situation of our partner retailer. Prior to our intervention, the retailer utilized a blanket completion-contingent

1
2
3 scheme (offering \$10 for any published review) combined with no targeted prosocial
4 reinforcement. This baseline approach was highly inefficient. While it engaged experienced
5 reviewers, it largely failed to activate new ones. Under this baseline condition, first-time
6 reviewers exhibited a mere 2.2% review response rate ($SD=14.7\%$). Furthermore, because the
7 reward was guaranteed regardless of review helpfulness, the resulting reviews were generally
8 short, leading to a cost of over \$200 per review that crossed the helpful-vote threshold. Our
9 results show that restructuring this budget can reduce the cost paid per helpful review from over
10 \$200 to less than \$40 (a reduction of roughly 80%) while simultaneously activating the
11 platform’s dormant first-time reviewer segment.
12
13
14
15
16
17
18
19
20
21
22

23
24 Our findings revealed that shifting from a completion-contingent scheme to a targeted
25 performance-contingent scheme can drastically improve these outcomes, particularly for first-
26 time reviewers who represent a large untapped resource for retailers. We find that retailers can
27 significantly increase review effort by restructuring their fixed budget, providing only 20% of the
28 incentive budget for completion, and reserving the remaining 80% as a performance-contingent
29 incentive triggered by a helpful vote. However, the efficacy of this performance-contingent
30 incentive depends heavily on the prosocial reinforcement used in the review solicitation email.
31
32
33
34
35
36
37
38
39
40

41 Table IV summarizes the expected response rates and review lengths for first-time
42 reviewers under each combination of incentive scheme and prosocial framing.
43
44
45

46 — Table IV around here —
47
48
49

50 As shown in Table IV, simply switching from the baseline to a targeted strategy that pairs
51 performance-contingent incentives with social exchange reinforcement (i.e., asking reviewers to
52 help other customers) causes the response rate among first-time reviewers to jump from 2.2% to
53
54
55
56
57
58
59
60

11.8% (H2a and H3a). This represents more than a five-fold increase in activating historically dormant reviewers without sacrificing review effort. Alternatively, if a manager's primary goal is maximizing review length from the first-time reviewer segment, pairing performance-contingent incentives with reciprocal exchange (i.e., asking reviewers to help the company) yields the longest average reviews (192.34 characters; consistent with H3b) while more than tripling the baseline response rate (7.0%).

For managers interested in targeted incentivization, we suggest a segmented approach. Because experienced reviewers show no significant effort response to performance-contingent incentives (H2b) and already participate at high rates under completion-contingent rewards, managers can maintain completion-contingent rewards for this group to maximize review volume. Conversely, managers should target first-time reviewers with performance-contingent incentives paired with strategic prosocial messaging to overcome their high barriers to entry and ensure high-quality initial contributions. Importantly, this reallocation requires no increase in the total incentive budget. It is a restructuring of the existing \$10 reward combined with a change in email copy, making it a low-cost intervention to implement and test.

Finally, we recognize that different platforms have different demographic makeups and strategic priorities. In a post-hoc analysis, we simulated the optimal strategies for maximizing conversion rates and the percentage of helpful reviews and minimizing the cost per helpful review. Based on these simulations, we provide a prescriptive decision tree (Figure 8).

— Figure 8 around here —

Figure 8 operationalizes these findings as a decision tool. A manager inputs two pieces of information their platform already tracks—the strategic objective (conversion volume, helpful

review share, or cost efficiency) and the percentage of first-time reviewers—and the tree returns the optimal incentive scheme (performance-contingent vs. completion-contingent) and prosocial reinforcement (social vs. reciprocal exchange). For example, a platform resembling our partner retailer’s composition (58% first-time reviewers) seeking cost-efficient helpful reviews should deploy performance-contingent incentives with social exchange framing.

Limitations and Future Research

In this research, we investigated the effect of performance-contingent incentives on reviewer effort for 1,704 unique products across 286 different brands. The demography was limited to the customers of one retailer. An additional limitation is that our primary outcome variable is review length rather than a direct measure of helpfulness. We made this choice for two main reasons. First, helpfulness votes are sparse on our retailer’s platform. As a result, helpfulness-based outcomes in this paper should be interpreted as exploratory reader-response measures. Second, our theoretical grounding explains the behavior of reviewers (i.e., their effort) when offered performance-contingent incentives, not how their behavior is perceived by third parties (i.e., helpfulness).

Furthermore, because we prioritized ecological validity by conducting a natural field experiment, we could not administer surveys to directly measure psychological mediators such as perceived reciprocity, autonomy, or competence. Future research could utilize controlled laboratory experiments to directly measure these underlying psychological mechanisms and formally test the mediated pathways implied by our behavioral findings.

Moreover, although our data consists of 1,704 unique products, future research should study the effectiveness of performance-contingent incentives across a wider range of products, brands, and services and on multiple retailers to further test the external validity of our findings.

Furthermore, we varied the amount of performance-contingent incentives as a proportion of a fixed total incentive of \$10 and found no effect of the proportion of performance-contingent incentives on reviewer effort. However, the absolute amount of an incentive can have an effect on reviewer effort (Falk, 2007; Gneezy and Rustichini, 2000). Therefore, future research could investigate the effect of varying the total incentive amount on the efficacy of performance-contingent incentives. Additionally, future research could investigate how the objective difficulty of the performance threshold (e.g., requiring a peer helpful vote versus a highly achievable system metric like a minimum word count) affects reviewer expectations and subsequent effort. In our study, the performance-contingent reward relies on a weak signal, as the vast majority of reviews never receive a helpful vote. This means many first-time reviewers who exerted effort under the performance-contingent condition ultimately did not receive the reward. Future research should investigate the longitudinal effects of this unmaterialized payoff. If reviewers exert effort but the reward proves unachievable, it may violate expectations and negatively impact future autonomous motivation.

Additionally, our managerial decision tree relies on a linear simulation of our empirical field data. While mathematically sound for independent actors within our sample, its broader application assumes that the observed effects of performance-contingency remain constant across platforms with vastly different macro-compositions. For instance, the baseline responsiveness of users on a mature platform may differ from users on a nascent platform, representing a boundary condition that future research could empirically explore.

Moreover, in our analysis, we did not account for various customer profiles and product types. Research has shown that online product reviews interact with different product types (Hong *et al.*, 2014). Future research should study the effect of the customer profile (e.g. impulsive buyers, Sojka and Giese, 2003), product type (e.g. search vs. experience), and their interaction with performance-contingent incentives to see whether different customer profiles respond differently to performance-contingent incentives and if the effect is moderated by product type.

Additionally, our field experiment observed reviewer behavior transitioning from a baseline of completion-contingent (guaranteed) incentives to performance-contingent incentives. Future research could investigate the reverse counterfactual: how reviewers respond when transitioning from performance-contingent to guaranteed incentives. Because a reviewer's baseline expectation establishes a reference point for perceived autonomy and competence, investigating how shifting incentive regimes interact with user history represents a promising avenue for understanding longitudinal review behavior.

Finally, while our descriptive data suggests that prosocial framing might alter the relationship between raw effort (length) and market usefulness (helpful votes), we did not formally test the semantic content of the reviews. Future research could utilize natural language processing to investigate whether reciprocal framing directs reviewer effort toward firm-related feedback (e.g., shipping, service) while social framing directs effort toward product-specific evaluations.

Ethical and Consent Statement

All procedures performed in studies involving human participants were conducted in adherence to national and international research ethics guidelines. Ethical approval for this research, including both the calibration studies and the field experiment, was obtained from the Institutional Review Board of [Institution Name withheld for peer review] (Approval Number: [Withheld]; Date of Approval: [Withheld]).

In accordance with relevant consent guidelines, informed consent was secured for all participants. For the calibration study, participants were informed of the research purpose, data storage procedures, and the protection of their anonymity, and explicit informed consent was obtained prior to participation. For field study, the field experiment was conducted in collaboration with a retail partner, adhering to privacy and consent protocols approved by the aforementioned Institutional Review Board to ensure that customer data and participant anonymity were strictly protected.

References

- Ai, J., Gursoy, D., Liu, Y. and Lv, X. (2022), "Effects of offering incentives for reviews on trust: Role of review quality and incentive source", *International Journal of Hospitality Management*, Elsevier, Vol. 100, p. 103101.
- Babić Rosario, A., De Valck, K. and Sotgiu, F. (2020), "Conceptualizing the electronic word-of-mouth process: What we know and need to know about eWOM creation, exposure, and evaluation", *Journal of the Academy of Marketing Science*, Springer, Vol. 48 No. 3, pp. 422–448.
- Baumeister, R.F. and Leary, M.R. (1995), "The need to belong: Desire for interpersonal attachments as a fundamental human motivation.", *Psychological Bulletin*, American Psychological Association, Vol. 117 No. 3, p. 497.
- Bénabou, R. and Tirole, J. (2004), "Incentives and Prosocial Behavior", *SSRN Electronic Journal*, doi: [10.2139/ssrn.639043](https://doi.org/10.2139/ssrn.639043).

- Box, G.E.P. and Cox, D.R. (1964), "An Analysis of Transformations", *Journal of the Royal Statistical Society: Series B (Methodological)*, Vol. 26 No. 2, pp. 211–243, doi: [10.1111/j.2517-6161.1964.tb00553.x](https://doi.org/10.1111/j.2517-6161.1964.tb00553.x).
- Burtch, G., Hong, Y., Bapna, R. and Griskevicius, V. (2018), "Stimulating online reviews by combining financial incentives and social norms", *Management Science*, INFORMS, Vol. 64 No. 5, pp. 2065–2082.
- Cabral, L. and Li, L. (2015), "A dollar for your thoughts: Feedback-conditional rebates on eBay", *Management Science*, INFORMS, Vol. 61 No. 9, pp. 2052–2063.
- Cadsby, C.B., Song, F. and Tapon, F. (2007), "Sorting and incentive effects of pay for performance: An experimental investigation", *Academy of Management Journal*, Academy of Management Briarcliff Manor, NY 10510, Vol. 50 No. 2, pp. 387–405.
- Chen, R., Yuan, R., Huang, B. and Liu, M.J. (2023), "Feeling warm or skeptical? An investigation into the effects of incentivized eWOM programs on customers' eWOM sharing intentions", *Journal of Business Research*, Elsevier, Vol. 167, p. 114178.
- Chen, Y., Zhang, Z., Zhou, J., Liu, C., Zhang, X. and Yu, T. (2023), "A cognitive evaluation and equity-based perspective of pay for performance on job performance: A meta-analysis and path model", *Frontiers in Psychology*, Frontiers Media SA, Vol. 13, p. 1039375.
- Chevalier, S. (2022), "U.S. Digital shoppers product review expectation by age 2019", Statista, 5 April, available at: <https://www.statista.com/statistics/1019495/online-shoppers-expectations-product-reviews-in-the-us/> (accessed 17 May 2023).
- Choi, H.S. and Maasberg, M. (2022), "An empirical analysis of experienced reviewers in online communities: What, how, and why to review", *Electronic Markets*, Springer, Vol. 32 No. 3, pp. 1293–1310.
- Deci, E.L., Koestner, R. and Ryan, R.M. (1999), "A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation.", *Psychological Bulletin*, American Psychological Association, Vol. 125 No. 6, p. 627.
- Dorner, V., Giamattei, M. and Greiff, M. (2020), "The Market for Reviews: Strategic Behavior of Online Product Reviewers with Monetary Incentives", *Schmalenbach Business Review*, Vol. 72 No. 3, pp. 397–435, doi: [10.1007/s41464-020-00094-y](https://doi.org/10.1007/s41464-020-00094-y).
- Duan, W., Gu, B. and Whinston, A.B. (2008), "Do online reviews matter? — An empirical investigation of panel data", *Decision Support Systems*, Vol. 45 No. 4, pp. 1007–1016, doi: [10.1016/j.dss.2008.04.001](https://doi.org/10.1016/j.dss.2008.04.001).
- Eisenberger, R., Rhoades, L. and Cameron, J. (1999), "Does pay for performance increase or decrease perceived self-determination and intrinsic motivation?", *Journal of Personality and Social Psychology*, American Psychological Association, Vol. 77 No. 5, p. 1026.
- Falk, A. (2007), "Gift Exchange in the Field", *Econometrica*, Vol. 75 No. 5, pp. 1501–1511, doi: [10.1111/j.1468-0262.2007.00800.x](https://doi.org/10.1111/j.1468-0262.2007.00800.x).

- Finkelstein, S.R. and Fishbach, A. (2012), "Tell me what i did wrong: Experts seek and respond to negative feedback", *Journal of Consumer Research*, University of Chicago Press Chicago, IL, Vol. 39 No. 1, pp. 22–38.
- Frey, B.S. and Jegen, R. (2001), "Motivation crowding theory", *Journal of Economic Surveys*, Wiley Online Library, Vol. 15 No. 5, pp. 589–611.
- Gagné, M. and Deci, E.L. (2005), "Self-determination theory and work motivation", *Journal of Organizational Behavior*, Wiley Online Library, Vol. 26 No. 4, pp. 331–362.
- Ghose, A. and Ipeirotis, P.G. (2010), "Estimating the helpfulness and economic impact of product reviews: Mining text and reviewer characteristics", *IEEE Transactions on Knowledge and Data Engineering*, IEEE, Vol. 23 No. 10, pp. 1498–1512.
- Gilal, F.G., Zhang, J., Paul, J. and Gilal, N.G. (2019), "The role of self-determination theory in marketing science: An integrative review and agenda for research", *European Management Journal*, Elsevier, Vol. 37 No. 1, pp. 29–44.
- Gneezy, U., Meier, S. and Rey-Biel, P. (2011), "When and why incentives (don't) work to modify behavior", *Journal of Economic Perspectives*, Vol. 25 No. 4, pp. 191–210.
- Gneezy, U. and Rustichini, A. (2000), "Pay enough or don't pay at all", *The Quarterly Journal of Economics*, MIT Press, Vol. 115 No. 3, pp. 791–810.
- Goes, P.B., Lin, M. and Au Yeung, C. (2014), "'Popularity effect' in user-generated content: Evidence from online product reviews", *Information Systems Research*, INFORMS, Vol. 25 No. 2, pp. 222–238.
- Goette, L. and Stutzer, A. (2020), "Blood donations and incentives: Evidence from a field experiment", *Journal of Economic Behavior & Organization*, Elsevier, Vol. 170, pp. 52–74.
- Greene, J. (2021), "14 Proven Ways to Encourage Customers to Write Reviews", Databox, 25 March, available at: <https://databox.com/how-to-encourage-customers-to-write-reviews> (accessed 17 May 2023).
- Hennig-Thurau, T., Gwinner, K.P., Walsh, G. and Gremler, D.D. (2004), "Electronic word-of-mouth via consumer-opinion platforms: What motivates consumers to articulate themselves on the internet?", *Journal of Interactive Marketing*, Elsevier, Vol. 18 No. 1, pp. 38–52.
- Hong, W., Chan, F.K.Y., Thong, J.Y.L., Chasalow, L.C. and Dhillon, G. (2014), "A framework and guidelines for context-specific theorizing in information systems research", *Information Systems Research*, INFORMS, Vol. 25 No. 1, pp. 111–136, doi: [10.1287/isre.2013.0501](https://doi.org/10.1287/isre.2013.0501).
- Huang, A.H., Chen, K., Yen, D.C. and Tran, T.P. (2015), "A study of factors that contribute to online review helpfulness", *Computers in Human Behavior*, Elsevier, Vol. 48, pp. 17–27.

- Jia, H., Shin, S. and Jiao, J. (2022), "Does the length of a review matter in perceived helpfulness? The moderating role of product experience", *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 16 No. 2, pp. 221–236.
- Khern-am-nuai, W., Kannan, K.N. and Ghasemkhani, H. (2018), "Extrinsic versus Intrinsic Rewards for Contributing Reviews in an Online Platform", *Information Systems Research*, doi: [10.1287/ISRE.2017.0750](https://doi.org/10.1287/ISRE.2017.0750).
- Kim, K. and Ahn, S.J. (2017), "Rewards that undermine customer loyalty? A motivational approach to loyalty programs", *Psychology & Marketing*, Wiley Online Library, Vol. 34 No. 9, pp. 842–852.
- Klein, N., Marinescu, I., Chamberlain, A. and Smart, M. (2018), "[Online Reviews Are Biased. Here's How to Fix Them](#)", *Harvard Business Review*, 6 March.
- Krishna, V., Shrestha, Y.R. and Krogh, G. von. (2024), "'The double-edged sword of inflated help': Unravelling the motivation crowding in community question-answering platforms", *Plos One*, Public Library of Science San Francisco, CA USA, Vol. 19 No. 3, p. e0297627.
- Lee, S. and Choeh, J.Y. (2020), "Using the Social Influence of Electronic Word-of-Mouth for Predicting Product Sales: The Moderating Effect of Review or Reviewer Helpfulness and Product Type", *Sustainability*, Multidisciplinary Digital Publishing Institute, Vol. 12 No. 19, p. 7952, doi: [10.3390/su12197952](https://doi.org/10.3390/su12197952).
- Li, H., Wang, X., Wang, S., Zhou, W. and Yang, Z. (2023), "The power of numbers: An examination of the relationship between numerical cues in online review comments and perceived review helpfulness", *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 17 No. 1, pp. 126–139.
- López, M., Sicilia, M. and Verlegh, P.W. (2022), "How to motivate opinion leaders to spread e-WoM on social media: Monetary vs non-monetary incentives", *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 16 No. 1, pp. 154–171.
- Mekler, E.D., Brühlmann, F., Tuch, A.N. and Opwis, K. (2017), "Towards understanding the effects of individual gamification elements on intrinsic motivation and performance", *Computers in Human Behavior*, Elsevier, Vol. 71, pp. 525–534.
- Moe, W.W. and Schweidel, D.A. (2012), "Online product opinions: Incidence, evaluation, and evolution", *Marketing Science*, INFORMS, Vol. 31 No. 3, pp. 372–386.
- Nguyen, L.T. and Hess, T.J. (2026), "The psychology of paid reviewers: An experimental study of online reviewer needs and motivations", *European Journal of Information Systems*, Taylor & Francis, pp. 1–30.
- Pitman, J. (2022), "Local Consumer Review Survey 2022: Customer Reviews and Behavior", BrightLocal, 26 January, available at: <https://www.brightlocal.com/research/local-consumer-review-survey/> (accessed 7 November 2022).

- Qiao, D., Lee, S.-Y., Whinston, A.B. and Wei, Q. (2020), "Financial incentives dampen altruism in online prosocial contributions: A study of online reviews", *Information Systems Research*, INFORMS, Vol. 31 No. 4, pp. 1361–1375.
- Qiao, D. and Rui, H. (2023), "Text performance on the vine stage? The effect of incentive on product review text quality", *Information Systems Research*, INFORMS, Vol. 34 No. 2, pp. 676–697.
- Racherla, P. and Friske, W. (2012), "Perceived 'usefulness' of online consumer reviews: An exploratory investigation across three services categories", *Electronic Commerce Research and Applications*, Elsevier, Vol. 11 No. 6, pp. 548–559.
- Rawal, M., Saavedra Torres, J.L., Bagherzadeh, R., Rani, S. and Melancon, J. (2024), "The role of individualism, collectivism, and promotional reward type on consumer response to amplified word-of-mouth strategies", *Journal of Consumer Marketing*, Emerald Publishing Limited, Vol. 41 No. 3, pp. 298–313.
- Rohde, C., Kupfer, A. and Zimmermann, S. (2022), "Explaining reviewing effort: Existing reviews as potential driver", *Electronic Markets*, Springer, Vol. 32 No. 3, pp. 1169–1185.
- Rummel, A. and Feinberg, R. (1988), "Cognitive evaluation theory: A meta-analytic review of the literature", *Social Behavior and Personality: An International Journal*, Scientific Journal Publishers, Vol. 16 No. 2, pp. 147–164.
- Ryan, R.M. and Deci, E.L. (2000), "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being.", *American Psychologist*, American Psychological Association, Vol. 55 No. 1, p. 68.
- Ryan, R.M., Mims, V. and Koestner, R. (1983), "Relation of reward contingency and interpersonal context to intrinsic motivation: A review and test using cognitive evaluation theory.", *Journal of Personality and Social Psychology*, American Psychological Association, Vol. 45 No. 4, p. 736.
- Smith, A. (2021), "How the Right Incentive Can Help You Generate More Reviews", PowerReviews, 4 August, available at: <https://www.powerreviews.com/blog/what-incentives-generate-reviews/> (accessed 17 May 2023).
- Sojka, J.Z. and Giese, J.L. (2003), "Using individual differences to detect customer shopping behaviour", *The International Review of Retail, Distribution and Consumer Research*, Routledge, Vol. 13 No. 4, pp. 337–353, doi: 10.1080/0959396032000129462.
- Vallerand, R.J. and Reid, G. (1984), "On the causal effects of perceived competence on intrinsic motivation: A test of cognitive evaluation theory", *Journal of Sport and Exercise Psychology*, Human Kinetics, Inc., Vol. 6 No. 1, pp. 94–102.
- Wadi, D., Legoux, R., Fredette, M. and Sénécal, S. (2026), "The interplay of altruism and financial incentives: Maximizing online reviews through effective messaging", *Journal of Electronic Commerce Research*, Vol. 27 No. 2.

Wang, C.L. (2024), “Editorial—what is an interactive marketing perspective and what are emerging research areas?”, *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 18 No. 2, pp. 161–165.

Wang, C.L. (2025), “Demonstrating contributions through storytelling”, *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 19 No. 1, pp. 1–4.

Wang, J., Ghose, A. and Ipeirotis, P. (2012), “Bonus, disclosure, and choice: What motivates the creation of high-quality paid reviews?”, *ICIS 2012 Proceedings*, presented at the International Conference on Information Systems 2012, Association for Information Systems, Orlando, FL, p. 9.

Wang, X. and Li, Y. (2014), “Trust, psychological need, and motivation to produce user-generated content: A self-determination perspective”, *Journal of Electronic Commerce Research*, California State University, Vol. 15 No. 3, pp. 241–253.

Weinstein, N. and Ryan, R.M. (2010), “When helping helps: Autonomous motivation for prosocial behavior and its influence on well-being for the helper and recipient.”, *Journal of Personality and Social Psychology*, American Psychological Association, Vol. 98 No. 2, p. 222.

Wooldridge, J.M. (2016), *Introductory Econometrics a Modern Approach*, South-Western cengage learning.

Woolley, K. and Sharif, M.A. (2021), “Incentives increase relative positivity of review content and enjoyment of review writing”, *Journal of Marketing Research*, SAGE Publications Sage CA: Los Angeles, CA, Vol. 58 No. 3, pp. 539–558.

Yin, D., Bond, S.D. and Zhang, H. (2017), “Keep Your Cool or Let it Out: Nonlinear Effects of Expressed Arousal on Perceptions of Consumer Reviews”, *Journal of Marketing Research*, SAGE Publications Inc, Vol. 54 No. 3, pp. 447–463, doi: [10.1509/jmr.13.0379](https://doi.org/10.1509/jmr.13.0379).

Yu, Y., Khern-am-nuai, W. and Pinsonneault, A. (2022), “When Paying for Reviews Pays Off: The Case of Performance-Contingent Monetary Rewards”, doi: [10.25300/MISQ/2022/15488](https://doi.org/10.25300/MISQ/2022/15488).

Zhang, Y., Niu, Y., Chen, Z., Deng, X., Wu, B. and Chen, Y. (2024), “Money matters? Effect of reward types on customers’ review behaviors”, *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 18 No. 3, pp. 391–409.

Zhu, J., Tse, D.K. and Fei, Q. (2018), “Effects of online consumer reviews on firm-based and expert-based communications”, *Journal of Research in Interactive Marketing*, Emerald Publishing Limited, Vol. 12 No. 1, pp. 45–78.

Figures

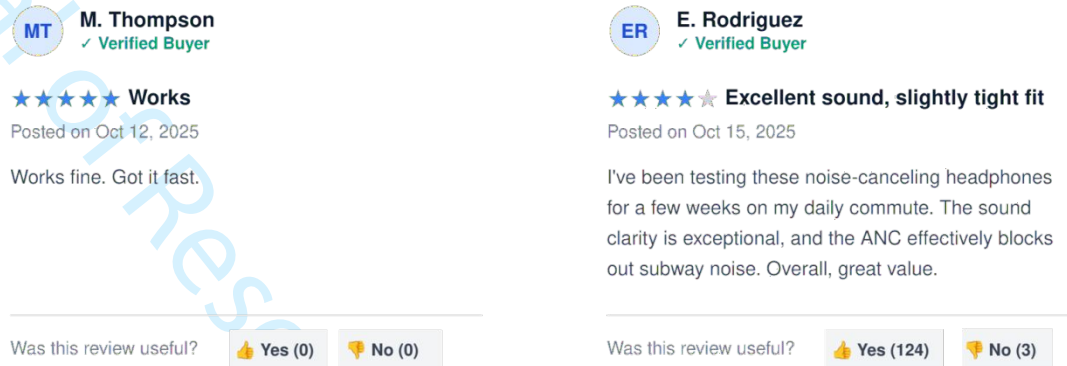


Figure 1: The managerial tension in online review generation: Retailers often receive short, unhelpful reviews (left) when they actually require longer, helpful reviews that provide value to consumers (right).

Source: Authors own work

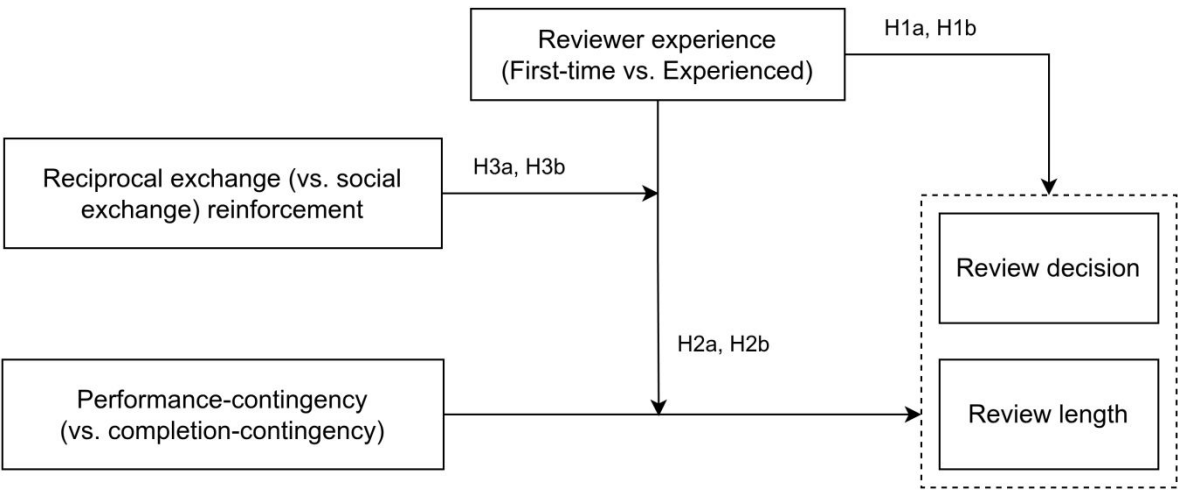


Figure 2: Theoretical framework

Source: Authors own work

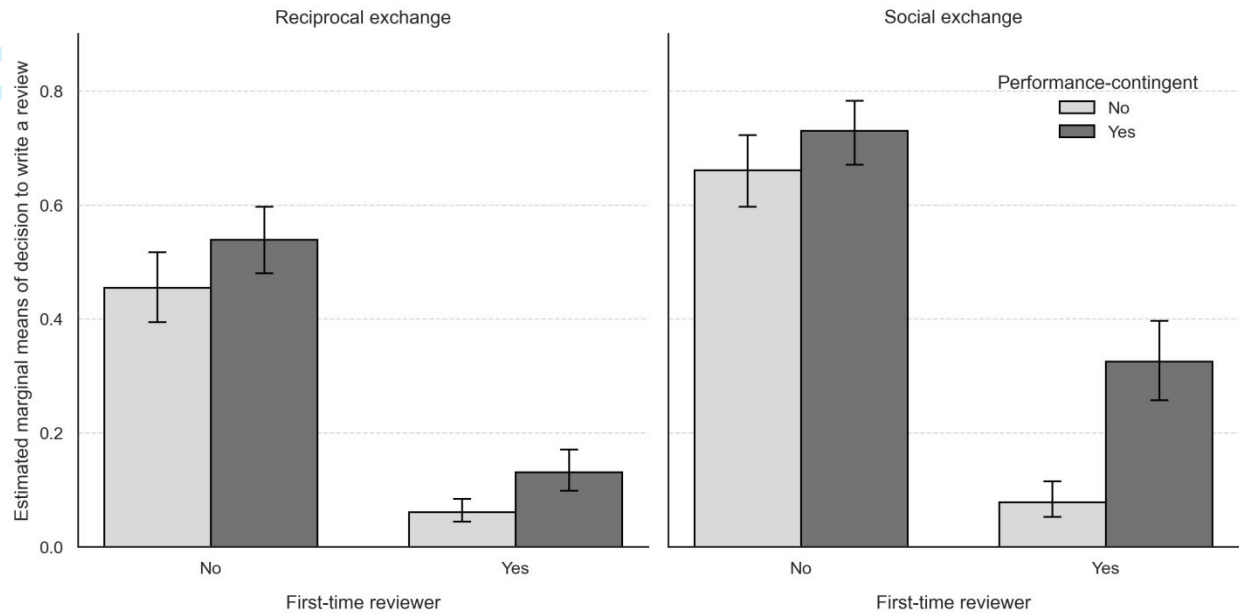


Figure 3: Estimated Marginal Means of Review decision for Reciprocal vs. Social exchange groups

Note. Error bars: 95% CI.

Source: Authors own work

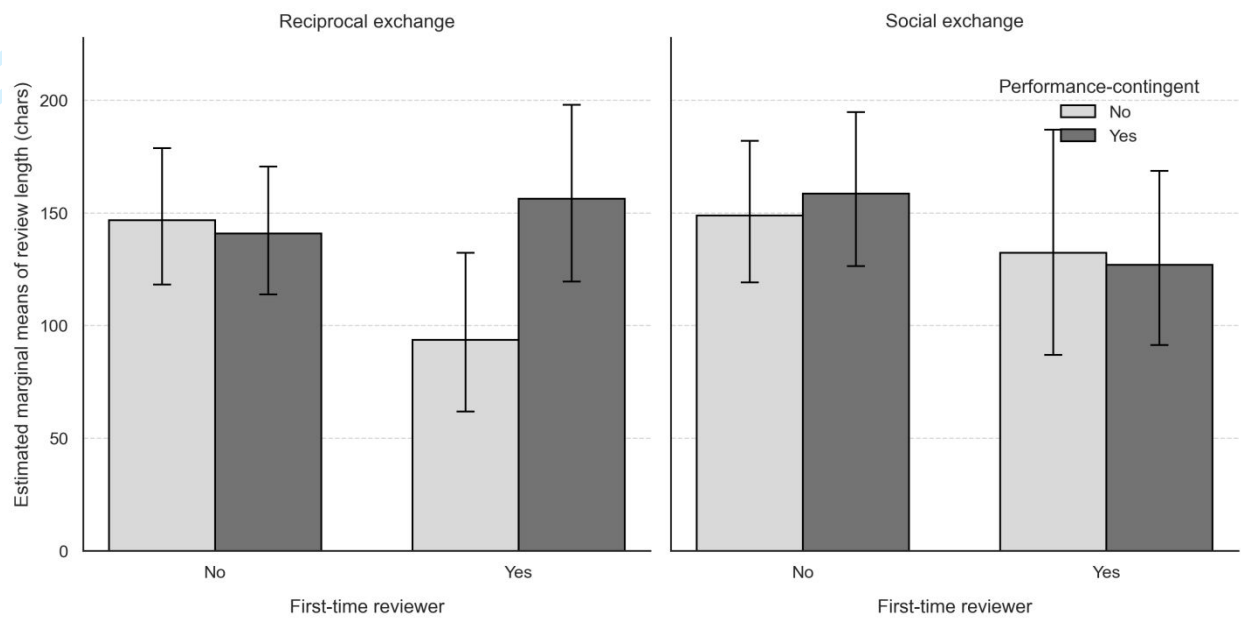


Figure 4: Estimated Marginal Means of Review length for Reciprocal vs. Social exchange groups

Note. Error bars: 95% CI.

Source: Authors own work

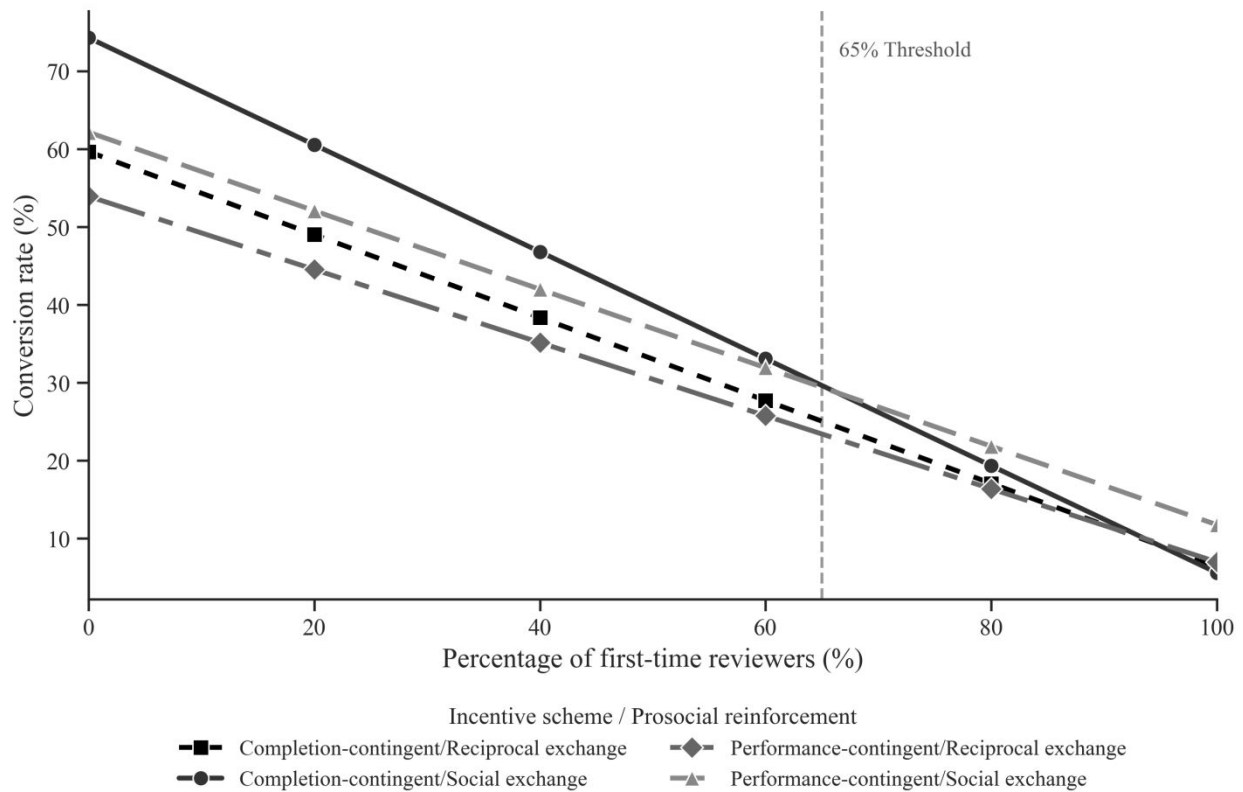


Figure 5: Conversion rate against first-time reviewer (%)

Source: Authors own work

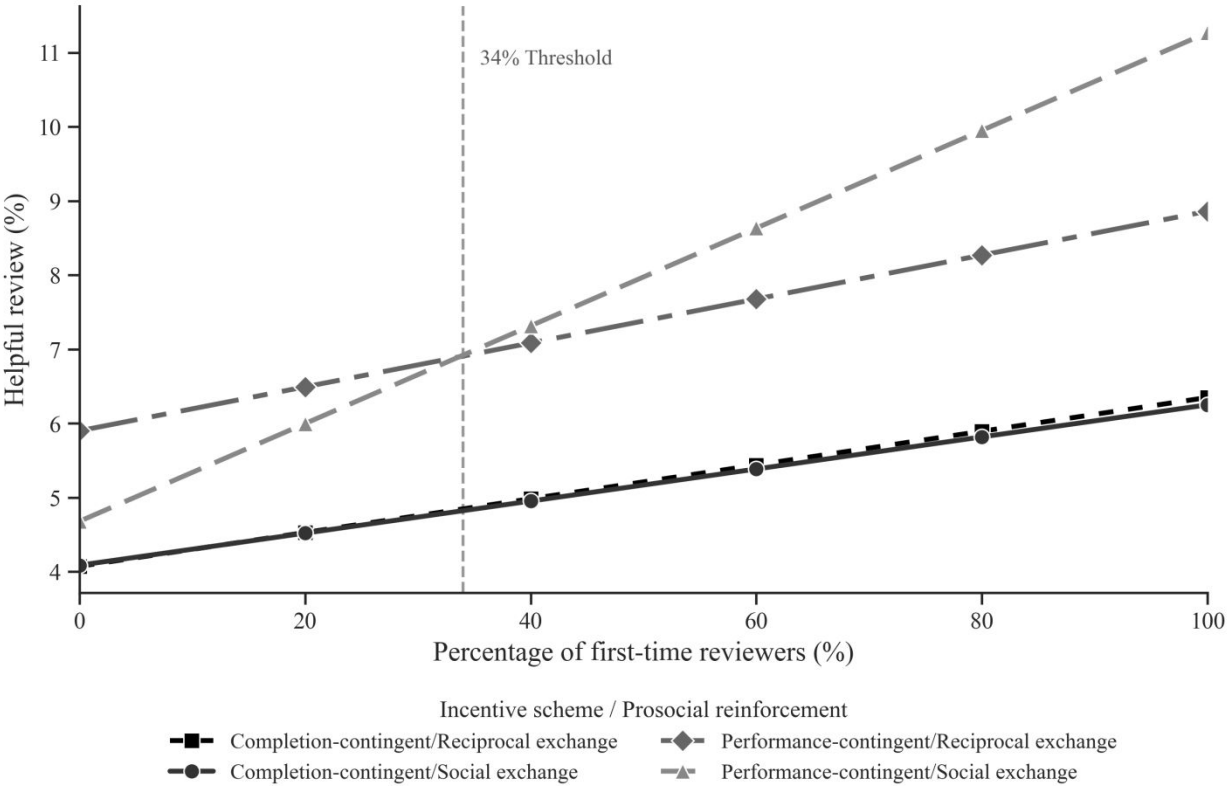


Figure 6: Helpful review (%) against first-time reviewer percentage

Source: Authors own work

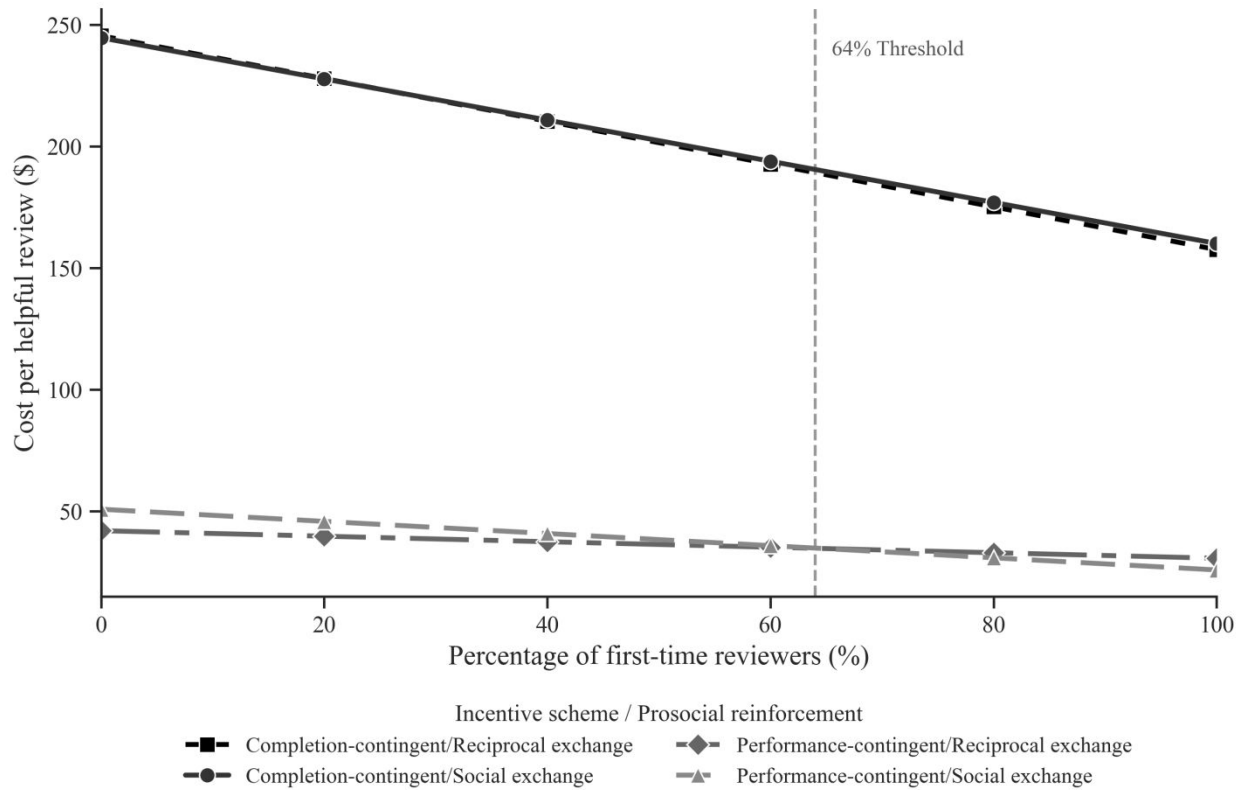


Figure 7: Cost per helpful review (\$) against first-time reviewer percentage

Source: Authors own work

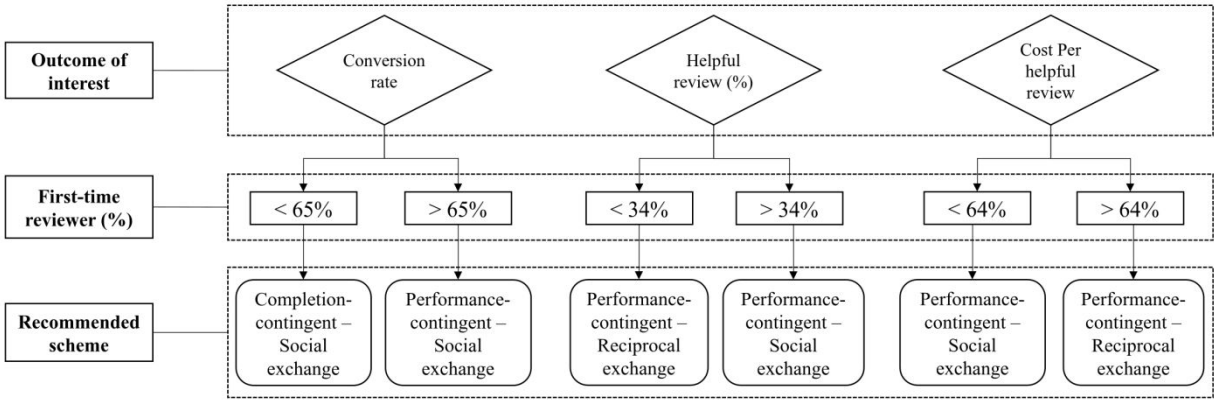


Figure 8: Decision tree for selecting the optimal incentive scheme and prosocial reinforcement based on the outcome of interest

Source: Authors own work

Tables

Table I: Data collection

Performance-contingency	Reciprocal exchange reinforcement	Email Title	Email Content	Emails sent	Valid emails ¹	Reviews written
Yes	Yes	Help us and receive \$10 credit	Hello {customerName}, We love to hear from you. Could you please take some time to write a review about {productName}? Your review helps us improve our business. You can receive a \$2 credit after your review of {productName} is published, and an \$8 credit after your review receives 1 helpful vote. Once published, you will receive an email confirming this fact, and the credit will be added directly to your account. Please note, the credit will expire in 90 days from the date of the confirmation email.	2923	1727	401
	No	Help other customers and receive \$10 credit	Hello {customerName}, Other customers love to hear from you. Could you please take some time to write a review about {productName}? Your review helps other customers, like you, make better purchase decisions. You	2113	981	306

can receive a \$2 credit after your review of {productName} is published, and an \$8 credit after your review receives 1 helpful vote. Once published, you will receive an email confirming this fact, and the credit will be added directly to your account. Please note, the credit will expire in 90 days from the date of the confirmation email.

No	Yes	Help us and receive \$10 credit	Hello {customerName}, We love to hear from you. Could you please take some time to write a review about {productName}? Your review helps us improve our business. You can receive a \$10 credit, valid on your next purchase, by reviewing {productName}. Once published, you will receive an email confirming this fact, and the \$10 credit will be added directly to your account. Please note, the credit will expire in 90 days from the date of the confirmation email.	2793	1884	603
	No	Help other customers and receive \$10 credit	Hello {customerName}, Other customers love to hear from you. Could you please take some time to write a review about {productName}? Your review helps other	1726	998	350

customers, like you, make better purchase decisions. You can receive a \$10 credit, valid on your next purchase, by reviewing {productName}. Once published, you will receive an email confirming this fact, and the credit will be added directly to your account. Please note, the credit will expire in 90 days from the date of the confirmation email.

Total	9555	5590	1660
-------	------	------	------

Note. ¹ Excluding the emails that lead to reviews following future reminders

Source: Authors own work

Table II: Descriptive statistics for review decision and length

Reviewer Type	Incentive Scheme	Prosocial Reinforcement	N	Review decision		Review length	
				M	SD	M	SD
Experienced	Completion-contingent	Reciprocal exchange	905	0.597	0.491	161.437	143.723
		Social exchange	428	0.743	0.437	159.572	141.402
	Performance-contingent	Reciprocal exchange	597	0.539	0.499	169.32	129.311
		Social exchange	378	0.622	0.486	180.945	169.378
First-time	Completion-contingent	Reciprocal exchange	979	0.064	0.246	109.984	141.98
		Social exchange	570	0.056	0.23	130.875	90.812
	Performance-contingent	Reciprocal exchange	1130	0.07	0.255	192.342	179.647
		Social exchange	603	0.118	0.323	153.535	184.929

Source: Authors own work

Table III: Logistic regression with review decision as criterion and multiple linear regressions with sqrt(Review length) as criterion

Variable	Decision			Length		
	Main	2-Way	3-Way	Main	2-Way	3-Way
Intercept	0.493** (0.155)	0.531** (0.169)	0.716*** (0.181)	14.960*** (1.007)	14.929*** (1.079)	14.885*** (1.078)
Performance-contingency	0.837*** (0.128)	0.716*** (0.189)	0.424* (0.210)	0.192 (0.506)	0.148 (0.695)	0.396 (0.702)
Reciprocal exchange reinforcement ^a	-1.058*** (0.094)	-0.983*** (0.133)	-1.117*** (0.142)	-0.181 (0.318)	-0.239 (0.396)	-0.087 (0.401)
First-time reviewer ^b	-2.908*** (0.085)	-3.482*** (0.171)	-3.896*** (0.227)	-0.907* (0.365)	-2.139** (0.745)	-0.700 (0.952)
Performance-contingency × Reciprocal exchange reinforcement		-0.320	0.021		-0.164	-0.644

	(0.203)	(0.229)	(0.725)	(0.750)
Performance-	0.920***	1.643***	1.570*	-0.628
contingency × First-				
time reviewer				
	(0.175)	(0.291)	(0.750)	(1.177)
First-time reviewer ×	0.186	0.821**	0.533	-1.735
Reciprocal exchange				
reinforcement				
	(0.180)	(0.273)	(0.747)	(1.197)
Performance-		-1.144**		3.698*
contingency × First-				
time reviewer ×				
Reciprocal exchange				
reinforcement				
		(0.360)		(1.527)
Review score		-0.668***	-0.668***	-0.678***
		(0.179)	(0.179)	(0.179)
Is popular brand		0.056	0.069	0.063

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46

Order day^c

Monday	0.558*** (0.166)	0.541*** (0.164)	0.439** (0.168)	-0.166 (0.612)	-0.045 (0.661)	-0.104 (0.661)
Saturday	2.246*** (0.194)	2.271*** (0.194)	2.169*** (0.196)	-0.294 (0.692)	-0.065 (0.713)	0.052 (0.714)
Sunday	-1.681*** (0.175)	-1.707*** (0.178)	-1.735*** (0.174)	0.117 (0.664)	0.131 (0.667)	0.206 (0.666)
Thursday	-1.278*** (0.192)	-1.134*** (0.191)	-1.163*** (0.192)	0.946 (0.812)	1.200 (0.874)	1.340 (0.874)
Tuesday	0.981*** (0.207)	1.123*** (0.204)	1.076*** (0.207)	1.116 (0.631)	1.204 (0.724)	1.209 (0.723)
Wednesday	0.075 (0.143)	0.322* (0.156)	0.299 (0.158)	0.329 (0.490)	0.421 (0.614)	0.462 (0.613)
—	—	—	—	—	—	—
Observations	5590	5590	5590	1660	1660	1660

(Pseudo) R-squared	0.359	0.363	0.365	0.020	0.022	0.026
--------------------	-------	-------	-------	-------	-------	-------

Note: *p<0.05; **p<0.01; ***p<0.001

Standard errors are reported in parentheses.

a. *Helping other customers* is set as the reference group.

b. *Experienced reviewer* is coded 0, and *First-time reviewer* is coded 1.

c. *Friday* is set as reference.

Source:

Authors

own

work

Table IV: Expected outcomes for first-time reviewers based on incentivization strategy based on our field data

Incentive scheme	Prosocial reinforcement	Response rate	Review length (Characters)
Completion-contingent (Baseline)	None	2.2% (SD=0.15)	147.30 (SD=154.62)
Completion-contingent	Social Exchange (Help Others)	5.6% (SD=0.23)	130.88 (SD=90.81)
Completion-contingent	Reciprocal Exchange (Help Company)	6.4% (SD=0.25)	109.98 (SD=141.98)
Performance-contingent	Reciprocal Exchange (Help Company)	7.0% (SD=0.26)	192.34 (SD=179.65)
Performance-contingent	Social Exchange (Help Others)	11.8% (SD=0.32)	153.54 (SD=184.93)

Source: Authors own work

MT

M. Thompson

✓ Verified Buyer

★★★★★ **Works**

Posted on Oct 12, 2025

Works fine. Got it fast.

Was this review useful?



Yes (0)



No (0)

ER

E. Rodriguez

✓ Verified Buyer

★★★★★ **Excellent sound, slightly tight fit**

Posted on Oct 15, 2025

I've been testing these noise-canceling headphones for a few weeks on my daily commute. The sound clarity is exceptional, and the ANC effectively blocks out subway noise. Overall, great value.

Was this review useful?



Yes (124)



No (3)

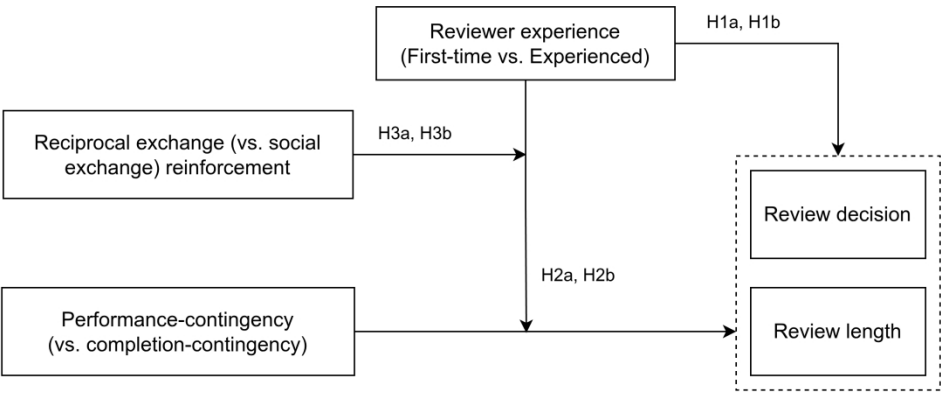


Figure 2: Theoretical framework

Source: Authors own work

966x433mm (72 x 72 DPI)

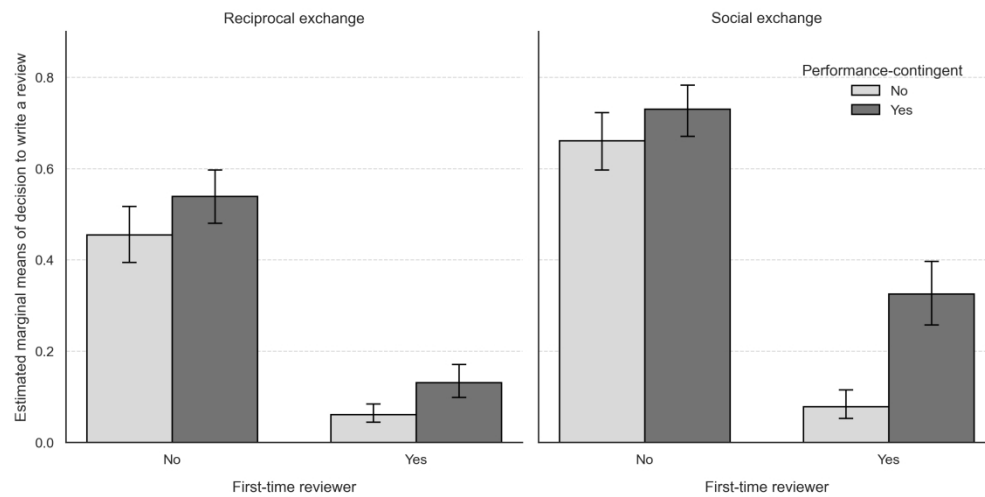


Figure 3: Estimated Marginal Means of Review decision for Reciprocal vs. Social exchange groups

Note. Error bars: 95% CI.

Source: Authors own work

628x315mm (118 x 118 DPI)

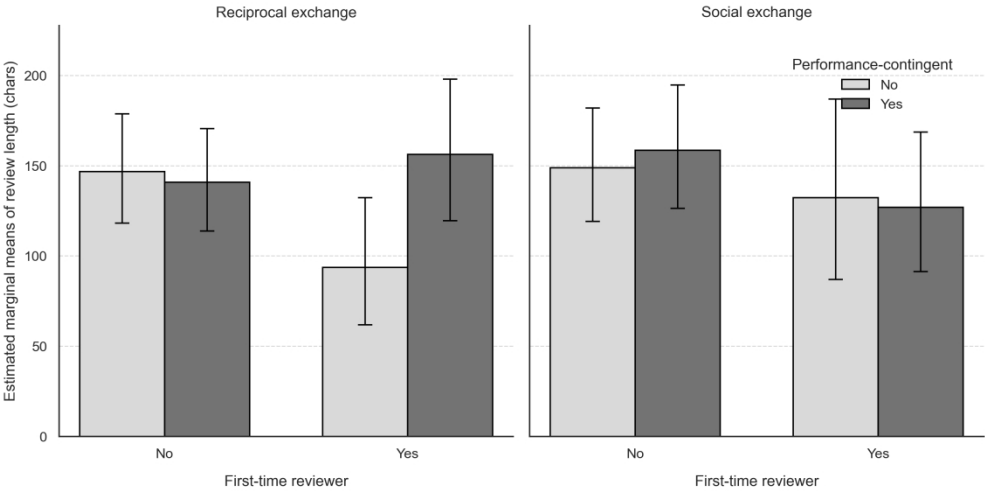


Figure 4: Estimated Marginal Means of Review length for Reciprocal vs. Social exchange groups

Note. Error bars: 95% CI.

Source: Authors own work

628x315mm (118 x 118 DPI)

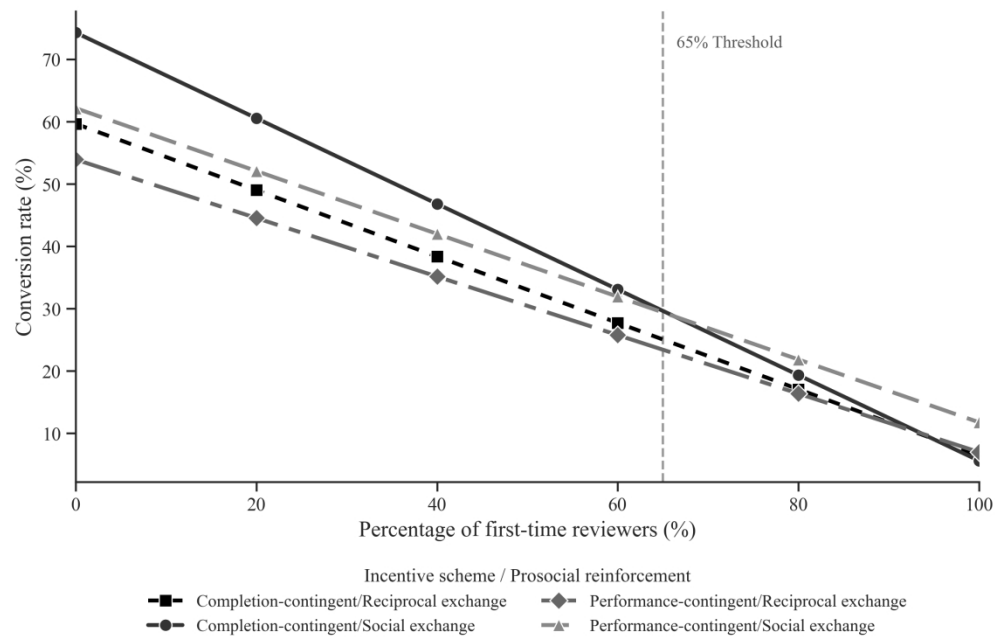


Figure 5: Conversion rate against first-time reviewer (%)

Source: Authors own work

570x372mm (118 x 118 DPI)

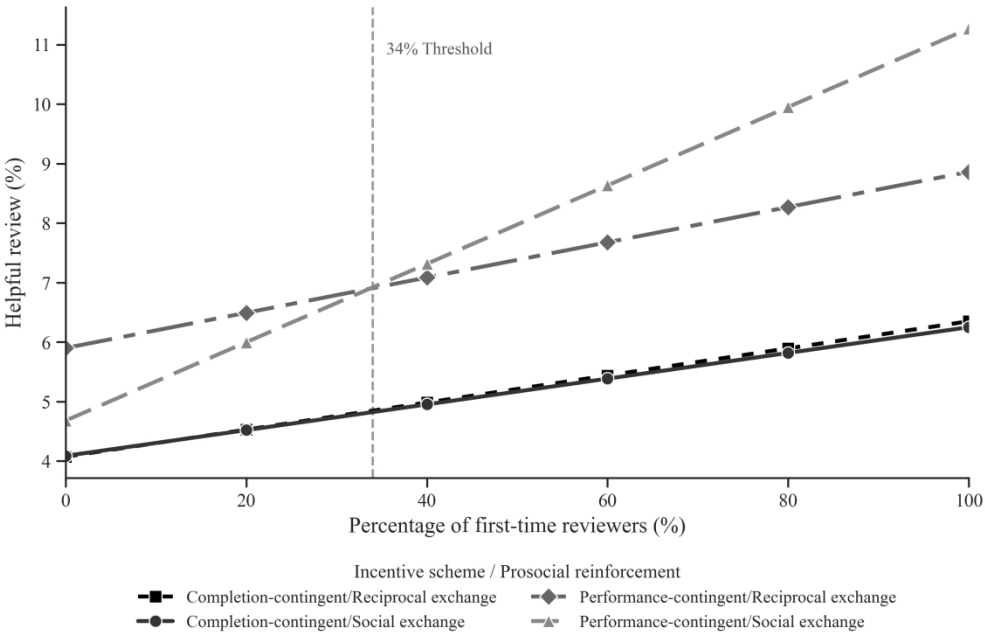


Figure 6: Helpful review (%) against first-time reviewer percentage

Source: Authors own work

570x372mm (118 x 118 DPI)

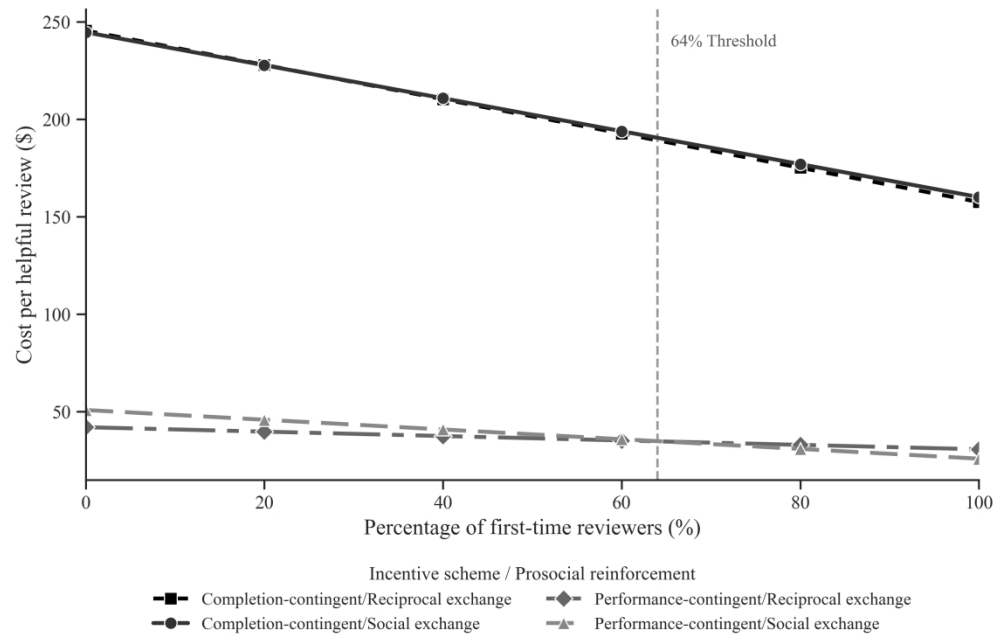


Figure 7: Cost per helpful review (\$) against first-time reviewer percentage

Source: Authors own work

570x372mm (118 x 118 DPI)

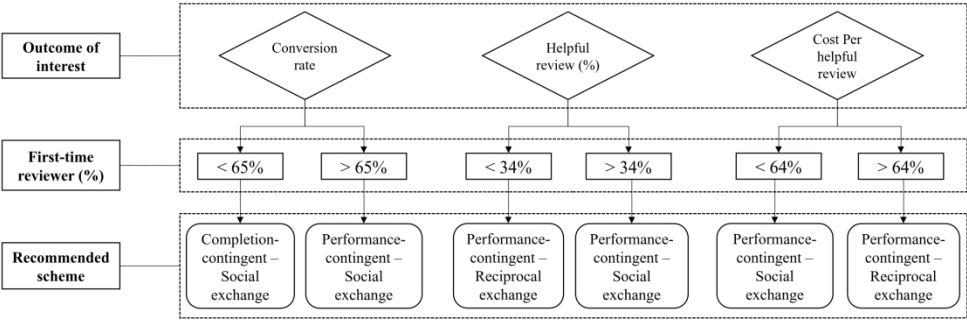


Figure 8: Decision tree for selecting the optimal incentive scheme and prosocial reinforcement based on the outcome of interestSource: Authors own work

810x290mm (76 x 76 DPI)

Web Appendix A

Experiment design

In this study, we calibrate the incentive structure for our main field experiment. Specifically, we investigate whether varying the percentage of the performance-contingent reward (vs. a completion-contingent reward) leads to differences in self-reported review effort. We also use this study to observe evidence for the directional patterns of behavior for first-time versus experienced reviewers before testing them in the field experiment.

We explore performance-contingent review incentivization in a one-factor (Performance-contingency percentage: 0% vs. 20% vs. 50% vs. 80% vs. 100%) between-subjects setup. [Table V](#) shows different performance-contingent incentive scheme levels and their corresponding message to the customers, given a maximum incentive of \$10.

Table V: Performance-contingency percentage of the groups and their corresponding review solicitation message

Contingent percentage	Message	Participants
0% (completion - contingent)	Could you please take some time to write a review about the product you purchased? You will receive \$10 credit after you submit your review.	23
20%	Could you please take some time to write a review about the product you purchased? You will receive \$8 credit after you submit your review and \$2 credit if your review receives 1 helpful vote .	21
50%	Could you please take some time to write a review about the product you purchased?	21

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

You will receive **\$5 credit** after you submit your review and **\$5 credit** if your review receives **1 helpful vote**.

80%

Could you please take some time to write a review about the product you purchased?

23

You will receive **\$2 credit** after you submit your review and **\$8 credit** if your review receives **1 helpful vote**.

100%

Could you please take some time to write a review about the product you purchased?

You will receive **\$10 credit** if your review receives **1 helpful vote**.

Source: Authors own work

Manipulation checks

To verify the manipulation of our treatment conditions in [Table V](#), we recruited 105 participants from Prolific. One participant did not finish the survey, and 3 failed the attention check. This left 101 valid participants ($M_{age}=37.63$, $SD=14.02$; 48.5% female). The participants were shown the product in [Figure 9](#). We ran a pretest to verify that the pair of gloves is gender neutral.



Figure 9: Product shown in the scenario of the calibration study

Source: Authors own work

After being shown the review solicitation message, the participants answered 4 items on a 7-point Likert scale Table VI, to check the manipulation of our 5 groups. Item 1 checks the manipulation of financial incentives. Since all groups are offered a type of financial incentive, all treatment groups must score above the scale's midpoint, i.e. 4. Item 2 checks the manipulation for the completion-contingent incentive group. Item 3 checks the manipulation of the

performance-contingent incentive groups. Finally, Item 4 checks the manipulation of the completion-contingent, 20% performance-contingent, and 50% performance-contingent groups versus the 80% performance-contingent and 100% performance-contingent groups. We run a one-tailed one-sample t-test for the first item and one-tailed independent samples t-tests for the rest of the items in Table VI to check the manipulation of our groups. The results of our manipulation check experiment show successful manipulation of the incentive and different levels of performance-contingent incentive. Table VI shows different groups, the expected manipulation check effects, and test results.

Table VI: Manipulation check items and results

Number	Item	Expected effect	<i>t</i>	<i>df</i>	<i>p</i>
1	I can receive a reward for writing the review.***	All groups $\geq$ 4	13.526	100	<0.001
2	I will receive a reward regardless of how helpful my review is.***	Completion-contingent group > other groups	10.482	46.536 ¹	<0.001
3	I need to write a helpful review to receive more rewards.***	Performance-contingent groups > completion-contingent group	8.266	99	<0.001
4	I will receive most of the rewards after my review receives a helpful vote.***	100% and 80% groups > 0%, 20%, and 50% groups	5.553	82	<0.001

Note. ¹ Equal variances not assumed.

*** $p < 0.001$.

Source: Authors own work

Sample and Procedure

We recruited 125 US participants from Prolific. The participants were limited to those who had a 95% or higher satisfaction rate. They received 1.50 USD for participating in the study. One participant did not finish the survey, and 15 participants failed the attention check, which explicitly asked them to select “Strongly disagree” for one question. This left us with 109 valid participants ($M_{age}=33.66$, $SD_{age}=12.24$; 47.7% female).

The participants were asked to imagine an online retailer which they have previously bought products from, saw the product image [Figure 9](#), and were shown the following scenario: “You ordered the product below [a pair of gloves] from their website. After your order is delivered, you receive the following email from the online retailer.” Then, they were shown one of the messages from [Table V](#), randomly assigned to each participant. After reading the email, the participants expressed their review intention and effort.

Measures

For the main experiment, after seeing the review solicitation email, the participants were asked their review intention (“Would you accept to write a review?”) on a 5-point Likert scale (adapted from Wadi *et al.*, 2026) and expected review effort (adapted from Yin *et al.*, 2017, items in Appendix). We used the arithmetic mean of the three items in the expected review effort to derive self-reported effort. Afterwards, we asked participants about their review experience (Wang *et al.*, 2012, items in Appendix) and manipulation check questions ([Table VI](#)).

We derived performance-contingent binary as a binary variable that takes the value 1 when the incentive scheme is performance-contingent and 0 when it is completion-contingent. We also derived first-time reviewer as a binary variable that takes the value 0 when a participant reports that they have previously written online product reviews (i.e., experienced reviewer) and 1 otherwise (i.e., first-time reviewer) to measure reviewer experience. Finally, we derived the performance-contingent proportion as a discrete variable that represents the proportion of the total incentive budget offered to the participants contingent on the performance (0.00 to 1.00).

Results

The 80% contingent group has the highest mean for review intention and the second-highest mean for self-reported effort. In comparison, the 100% contingent group has the highest mean for self-reported effort (Table VII).

Table VII: Means and standard deviations for Review intention and self-reported Effort

Performance-contingent percentage	Review intention		Self-reported effort	
	M	SD	M	SD
0	4.09	1.00	6.27	1.94
20	3.90	1.04	6.68	1.84
50	4.10	1.18	5.90	2.06
80	4.43	0.79	6.74	1.70
100	3.81	1.12	6.82	1.35
Total	4.07	1.03	6.49	1.79

Note. M and SD represent mean and standard deviation, respectively.

Source: Authors own work

The descriptive data provides initial directional patterns predicted for reviewer experience (Figure 10 and Figure 11). Consistent with our expectations, experienced reviewers generally reported higher baseline intentions to write a review (H1a) and greater expected effort (H1b) than first-time reviewers across conditions.

We also observe the expected interaction pattern (H2). For experienced reviewers, the introduction of a performance-contingent incentive does not show a clear increase in self-reported review intention or effort compared to a guaranteed completion reward. However, for first-time reviewers, moving from a completion-contingent reward to a performance-contingent reward shows a clear directional increase in self-reported effort.

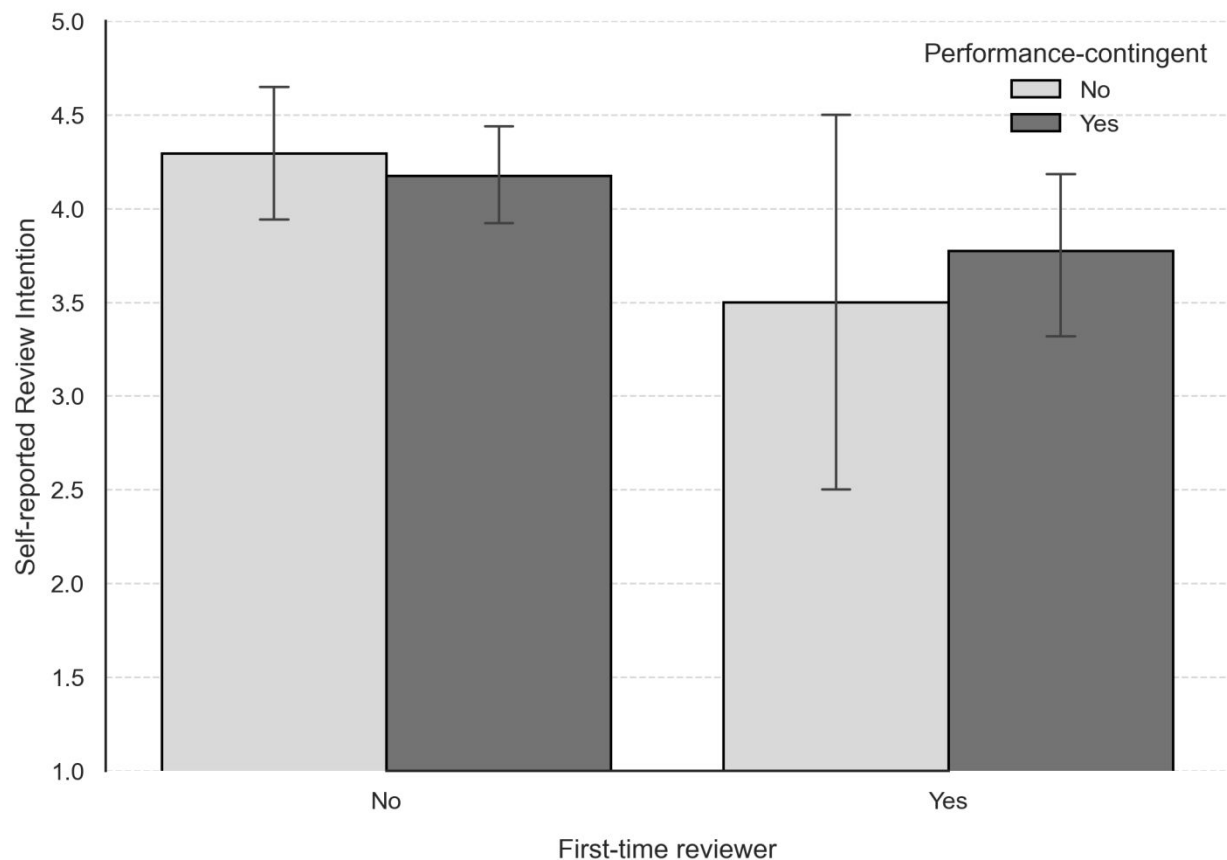


Figure 10: Means of self-reported review intention

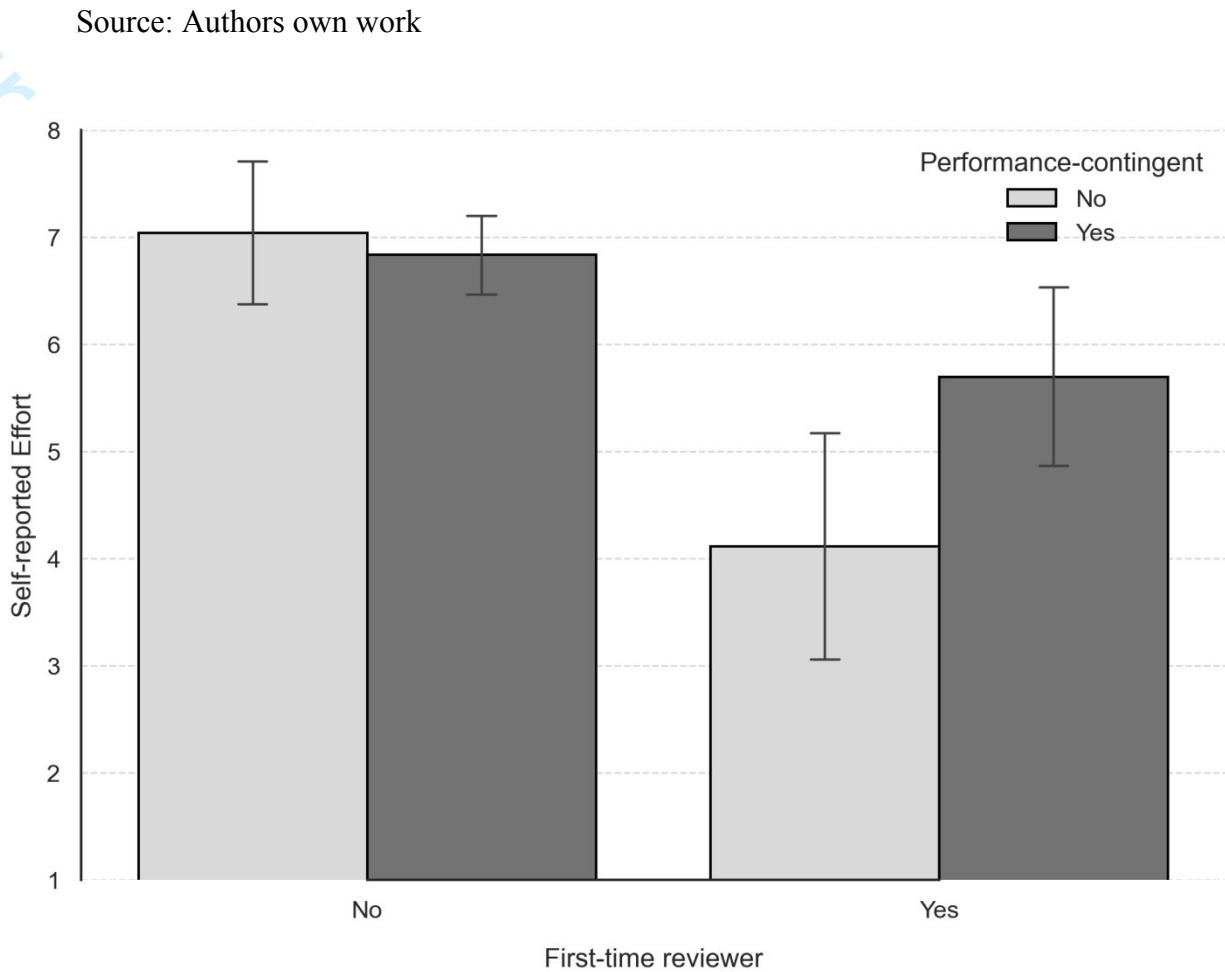


Figure 11: Means of self-reported review effort

Note. Error bars: 95% CI.

Source: Authors own work

Discussion

The results of the calibration study show that the exact proportion of the performance-contingent incentive (e.g., 20% vs. 80%) does not significantly alter self-reported effort, suggesting that the presence of the performance-contingent incentive itself is the primary driver of behavior. Moreover, the effect of the performance-contingent amount on review effort is not

significant but shows a positive pattern. Specifically, we see that 80% performance-contingent incentive has the highest average review intention ($M = 4.43$, $SD = 0.79$) and the second-highest self-reported effort ($M = 6.74$, $SD = 1.70$; Table VII). This pattern suggests that providing a small percentage (20% in this study) of the incentive budget for the completion of the task and the remaining budget (80%) after the review receives a helpful vote could lead to higher review behavior while saving the retailer a large proportion of the budget in case of no helpful votes. Because the 80% performance-contingent condition yielded the highest directional intention and second-highest effort, we select this ratio for our field intervention.

Furthermore, this study provides initial directional patterns claimed by the hypotheses. There is a pattern indicating lower self-reported review intentions and effort for performance-contingent incentives and higher self-reported review intention and effort for experienced reviewers (vs. first-time reviews). There is also a pattern indicating stronger response to performance-contingent incentives by first-time reviewers.

Scales

- Review intention (Adapted from Wadi *et al.*, 2026)
 - Would you accept to write a review? (1= “definitely not”, 5 = “definitely yes”)
- Expected review effort (Adapted from Yin *et al.*, 2017)
 - If you write a review,
 - How much effort would you put into writing this review?
 - How much thought would you give to this review?
 - How much time would you spend writing this review?
- Unpaid and paid review experience (Wang *et al.*, 2012)
 - Have you written a review online before? (Yes/No)
 - Have you been paid to write a review on any website before? (Yes/No)

Web Appendix B

Field experiment - Manipulation Checks

To verify the manipulation of the social versus reciprocal exchange framings used in the review solicitation emails, we conducted an independent manipulation check study. We recruited 120 US participants from Prolific. Three participants failed an attention check that explicitly asked them to select “Strongly disagree,” leaving 117 valid participants ($M_{age}=31.92$; 45% female).

Participants were randomly assigned to read either the “Help other customers” email (social exchange framing, $n=60$) or the “Help us” email (reciprocal exchange framing, $n=57$). After reading the email, participants indicated their agreement with two statements on a 5-point Likert scale to measure their perceived motives: (1) “I can help others buy the right product” (measuring social exchange), and (2) “I am so satisfied with the company and its product that I want to help the company to be successful” (measuring reciprocal exchange).

We conducted independent samples Welch’s t-tests (equal variances not assumed) to compare the means between the two groups. As expected, participants in the social exchange condition reported significantly higher agreement with the statement “I can help others buy the right product” ($M=4.28$, $SD=0.80$) compared to those in the reciprocal exchange condition ($M=3.14$, $SD=1.25$); $t(99.6)=5.86$, $p<.001$. Conversely, participants in the reciprocal exchange condition reported significantly higher agreement with the statement about wanting to help the company be successful ($M=3.58$, $SD=0.92$) compared to those in the social exchange condition ($M=2.68$, $SD=1.37$); $t(103.5)=-4.16$, $p<.001$. These results confirm that the review

solicitation emails successfully manipulated the intended prosocial motives. Figure 12 shows the agreement scores for the two items.

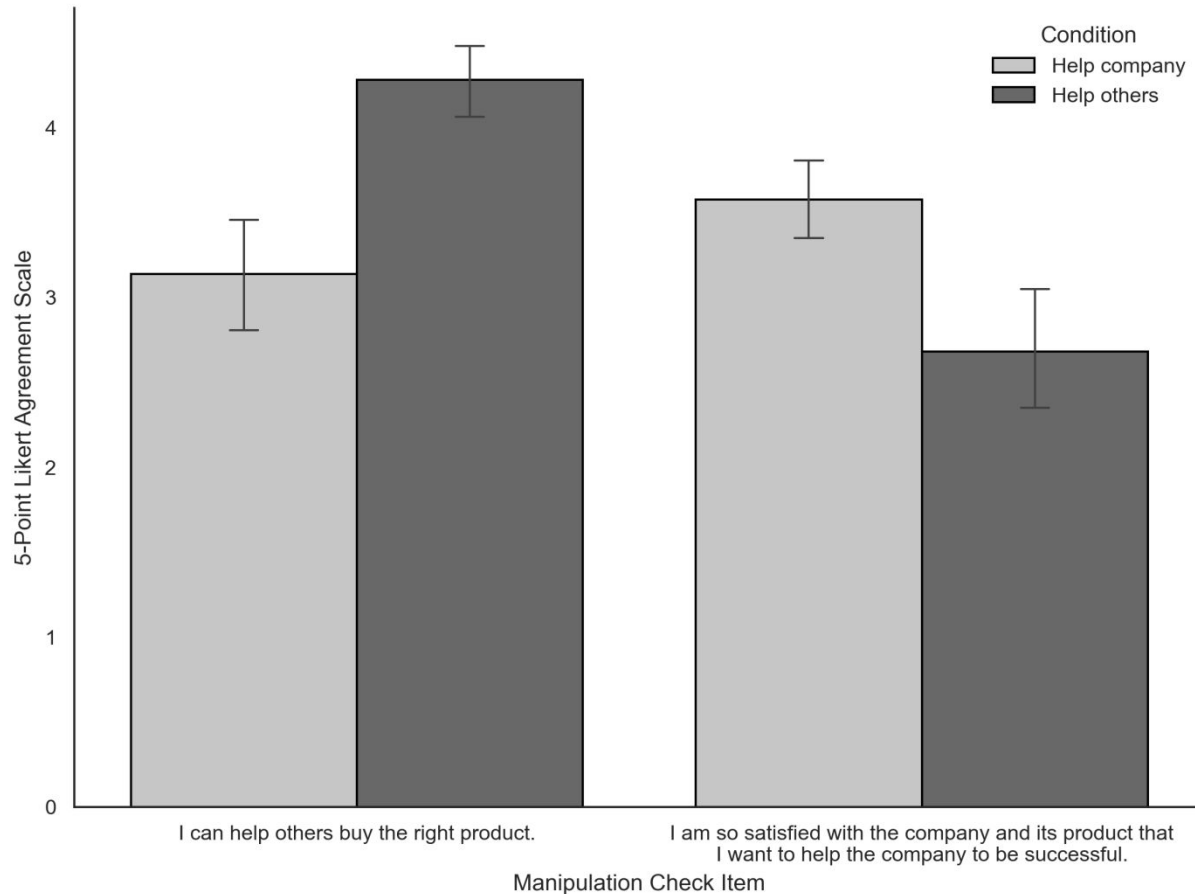


Figure 12: Manipulation check

Web Appendix C

Randomization and Balance Checks

To address potential endogeneity concerns arising from the review solicitation emails being dispatched in different time windows, we conducted a randomization check on key pre-

treatment covariates. We compared product-level and user-level characteristics across the four experimental conditions to ensure no systematic, unobserved differences drove our results.

Product-related covariates were well balanced. An ANOVA on the product rating (*ReviewScore*) showed no significant differences across the conditions ($p=0.524$; see Table VIII). Similarly, a Chi-Square test of independence on *brand_type* (popular vs. niche products) showed no significant variations in the product mix across the days the emails were sent ($p=0.204$; see Table IX).

We observed a statistically significant variation in the proportion of first-time reviewers ($p<.001$). However, because reviewer experience (*first-time reviewer*) is a central moderating construct in our theoretical framework (H2-H3), it is explicitly incorporated into all our regression models. Consequently, any baseline heterogeneity in the proportion of first-time reviewers across the time windows is mathematically accounted for, allowing us to accurately isolate the causal interaction effects of our experimental manipulations.

Table VIII: Balance Check for Continuous Pre-Treatment Covariates across Conditions

Incentive Scheme	Prosocial Reinforcement	Review Score (1-5)	
		<i>M</i>	<i>SD</i>
Completion-contingent	Reciprocal exchange	4.67	0.94
	Social exchange	4.73	0.81
Performance-contingent	Reciprocal exchange	4.71	0.88
	Social exchange	4.71	0.82
Test Statistic		$F=0.747$	
<i>p</i> -value		0.524	

Table IX: Balance Check for Categorical Pre-Treatment Covariates across Conditions

Incentive Scheme	Prosocial Reinforcement	First-time Reviewer (%)	Popular Brand (%)
Completion-contingent	Reciprocal exchange	52.0%	52.7%
	Social exchange	57.1%	49.4%
Performance-contingent	Reciprocal exchange	65.4%	47.1%
	Social exchange	61.5%	46.4%
Test Statistic		$\chi^2 = 71.682$	$\chi^2 = 4.599$
p-value		< 0.001	0.204

Web Appendix D

To verify the robustness of our findings and validate our operationalization of review effort, we conducted secondary analyses using review helpfulness as the dependent variable. Helpfulness was operationalized as a binary indicator (*helpfulBin*) taking the value of 1 if a review received at least one helpful vote within 90 days, and 0 otherwise.

It is important to note a severe methodological constraint: helpful votes are highly sparse in this live retail environment. Only 5.2% of the reviews generated during our experiment received a helpful vote. Because 94.8% of the observations are zeros, standard regression models suffer from a lack of statistical power, making it difficult to detect complex (e.g., three-way) interaction effects. Therefore, these models are presented as exploratory robustness checks rather than primary causal tests.

D1. Validating Review Length as a Proxy for Usefulness

First, to address the concern that longer reviews might simply reflect verbosity rather than actual usefulness, we ran a logistic regression predicting *helpfulBin* using the square root of Review Length as the primary predictor. We included product-level covariates (Review Score, Brand Popularity) and day-of-week fixed effects to control for baseline differences.

Table X: Logistic regression results with Binary Helpfulness (*helpfulBin*) as criterion

Variable	Model D1: Validation	Model D2: Treatment Effects
Intercept	-4.067*** (0.894)	-2.342* (0.958)
Square root of Review length	0.088*** (0.019)	
Performance contingency		-0.529 (0.642)
Reciprocal exchange reinforcement		-0.333 (0.410)
<i>a</i>		
First-time reviewer ^b		0.370 (0.789)
Performance contingency × Reciprocal exchange reinforcement		1.210 † (0.645)
Performance contingency × First- time reviewer		0.630

		(0.928)
First-time reviewer × Reciprocal		-0.210
exchange reinforcement		(0.988)
Performance contingency × First-		-0.342
time reviewer × Reciprocal		
exchange reinforcement		(1.205)
Review score	0.092	-0.016
	(0.168)	(0.160)
Is popular brand	0.292	0.257
	(0.228)	(0.226)
Order day^c		
Monday	-0.858**	-0.823
	(0.302)	(0.563)
Saturday	-0.406	-0.272
	(0.331)	(0.579)
Sunday	-1.267 †	-1.266 †
	(0.752)	(0.755)
Thursday	0.072	-0.164
	(0.523)	(0.600)
Tuesday	-0.954 †	-1.344*
	(0.555)	(0.621)
Wednesday	-0.853*	-1.232*

	(0.389)	(0.478)
—	—	—
Observations	1660	1660
Pseudo R-squared	0.051	0.035

Note: † p<0.10; *p<0.05; **p<0.01; ***p<0.001

Standard errors are reported in parentheses. Empty cells indicate the variable was not included in that specific model.

a. *Helping other customers* is set as the reference group.

b. *Experienced reviewer* is coded 0, and *First-time reviewer* is coded 1.

c. *Friday* is set as reference.

The results confirm a significant, positive relationship between review length and the probability of receiving a helpful vote ($B=0.088$, $SE=0.019$, $z=4.62$, $p<.001$; Table X). This empirical finding validates our methodological approach. While length directly measures the writer’s *effort*, reviews into which users invest more effort are indeed evaluated as significantly more *useful* by the peer consumer market.

D2. Treatment Effects on Helpfulness Outcomes

Second, we replicated our primary 2x2x2 experimental model using *helpfulBin* as the dependent variable via logistic regression. The model included Performance contingency, Reciprocal exchange reinforcement, First-time reviewer, and their interactions, alongside our standard covariates.

As expected, given the extreme class imbalance, the overall model fit was marginal (LLR $p=.068$), and the complex three-way interaction did not reach statistical significance ($B=-0.34$, $SE=1.20$, $p=.777$).

Ultimately, this robustness check confirms two critical points: (1) review length is a highly valid, statistically robust proxy for predicting market usefulness, and (2) the severe sparsity of actual helpfulness votes renders it unsuitable as the primary dependent variable for testing complex causal interactions in a field setting, validating our primary reliance on review length.